# S2DBFT: Spectral–Spatial Dual-Branch Fusion Transformer for Hyperspectral Image Classification

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***Abstract*—Convolutional neural network (CNN)- and Transformer-based models have achieved remarkable success in hyperspectral image (HSI) classification tasks due to their outstanding ability to extract spatial and spectral features. However, most existing methods process spatial and spectral features separately, making it difficult to effectively learn their interactive features. To address this issue, we propose a spectral–spatial dual-branch fusion Transformer (S2DBFT) for HSI classification. Initially, we construct a spectral feature extraction module (SPEEM) and a spatial feature extraction module (SPAEM) to extract low-level features. These two modules consist of a 1-D convolution layer and a 2-D convolution layer, performing shallow extraction of spectral and spatial features. Next, the two feature sets obtained are fused through a weighted fusion process. Additionally, we design a multihead spectral–spatial self-attention (MHS3A) mechanism to enhance the interactive fusion of spectral and spatial features. Upon completion of feature fusion, a linear layer is used to obtain the sample labels. Extensive experiments on four HSI datasets demonstrate the effectiveness of the proposed S2DBFT, compared to existing state-of-the-art methods. In terms of performance evaluation, the overall accuracy (OA) and average accuracy (AA) indicate the superiority and generalizability of S2DBFT.**

***Index Terms*—Dual-branch fusion, hyperspectral image (HSI) classification, multihead spectral–spatial self-attention (MHS3A), Transformer.**

## I. Introduction

HYPERSPECTRAL images (HSIs) have witnessed remarkable enhancements in spatial and spectral resolution due to the swift progress of spectral imaging technology, facilitating the acquisition of 2-D spatial data coupled with 1-D spectral information, which collectively encapsulate a plethora of intricate details [1]. Due to the fine spectral characteristics, HSIs have found widespread applications across various fields, including precision (PR) agriculture [2], biomedical imaging [3], urban planning [4], mineral exploration [5], and environmental monitoring [6], [7], [8]. The core tasks in these applications often involve processes such as data denoising [9], spectral unmixing [10], detection [11], and classification [12], [13]. Among these techniques, land-cover classification is of particular importance, as it aims to assign appropriate class labels to each pixel based on both the spectral and spatial features present within the image. However, the high dimensionality and redundancy of HSI data, combined with the limited availability of labeled samples, pose significant challenges in model training, making it difficult to accurately identify different land-cover categories and thus impacting overall classification accuracy [14]. Consequently, one of the key challenges in current HSI classification techniques is the extraction of effective features from these high-dimensional datasets, as well as addressing the issue of sparse sample labels.

In the early stages of HSI classification (HSIC) research, traditional methods such as polynomial regression [15], Bayesian estimation [16], linear regression [9], support vector machines [10], maximum a posteriori estimation [11], random forests [17], and principal component analysis (PCA) [18] were widely used. These approaches typically relied on manually designed features and focused primarily on spectral information. However, they faced challenges related to feature extraction, high-dimensional data management, and the modeling of nonlinear relationships [14]. In recent years, the rapid development of deep learning technologies, particularly their exceptional performance in feature extraction, has garnered widespread attention and become a central focus of research. These advancements have led to significant breakthroughs in computer vision tasks, including image classification [19], object detection [20], and image segmentation [21], showcasing their powerful potential. Despite the remarkable advancements made by deep learning in HSI classification, the high dimensionality and limited sample size inherent in HSI data continue to present significant challenges. While convolutional neural networks (CNNs) [22] effectively capture local features, they struggle with long-range dependencies.

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Transformer-based models offer strong global modeling but often lack local sensitivity. Prior works have explored dual-branch networks to fuse spectral and spatial information. Gao et al. [23] used multiscale attention, while Alkhatib et al. [24] combined real and complex-valued networks for spatial and frequency modeling. Song et al. [25] proposed BS2T using multihead attention. However, limitations remain in interaction modeling, shallow fusion, and unbalanced feature weighting. To address these issues, we propose a spectral–spatial dual-branch fusion Transformer (S2DBFT), which employs hierarchical weighting and interaction-aware fusion to enhance feature representation and classification accuracy. Initially, dimensionality reduction is achieved via PCA. We then design the spatial feature extraction module (SPAEM) and spectral feature extraction module (SPEEM) to capture shallow spatial–spectral features. A spectral attention module (SAM), incorporating convolution and rectified linear unit (RELU), is employed to weight the features from SPAEM. To enhance feature interaction, we introduce multihead spectral–spatial self-attention (MHS3A) in place of traditional multihead self-attention (MHSA), yielding spectral–spatial interactive features. These shallow spatial–spectral features, interactive features, and the weighted spectral features are subsequently adaptively fused and classified using global average pooling (GAP) and a fully connected layer.

The main contributions of this work are summarized as follows.

1) We propose the S2DBFT method, which involves designing feature extraction modules to extract shallow spatial–spectral features, including the SPAEM and SPEEM, composed of 2-D CNN layers and 1-D CNN layers, respectively.
2) We design the SAM to weight the extracted spectral features, making them suitable for the subsequent feature interaction process.
3) We also design the multihead spatial–spectral self-attention module to achieve interactive fusion of spatial–spectral features and fuse the resulting spatial–spectral interactive features with the spectral features output by SAM.
4) Experimental results on four benchmark datasets demonstrate that the proposed S2DBFT outperforms other state-of-the-art methods in terms of classification performance.

The remainder of this article is as follows. Section II reviews the relevant research. Section III provides a comprehensive explanation of the proposed S2DBFT method. Section IV presents and analyzes the experimental results across four datasets. Section V further examines the clustering performance of the proposed approach. Section VI concludes with a summary of our contributions and outlines potential directions for future research.

## II. Related Work

CNN-based methods have become the dominant approach for HSI classification due to their strong performance in spatial feature extraction [26], [27], [28], [29]. Initially, CNNs such as 1-D-CNN [30], 2-D-CNN [31], and 3-D-CNN [32] were used for this task. Although 2-D-CNNs facilitated automated feature extraction, the numerous spectral bands in HSIs resulted in an increase in algorithm parameters. To address this, 3-D-CNNs were introduced. Zhong et al. [33] developed SSRN, an end-to-end 3-D-CNN that uses spectral and spatial residual units for feature extraction, with batch normalization (BN) enhancing performance. Roy et al. [34] proposed HybridSN, which combines 3-D-CNN and 2-D-CNN to capture joint spectral–spatial features and refine high-level spatial features. Li et al. [35] introduced a dual-branch dual-attention mechanism (DBDA) that uses 3-D-CNN to capture spectral and spatial features separately, optimizing both through attention blocks, improving feature extraction accuracy, particularly with limited training data.

Recently, the Transformer model has emerged as a significant research direction in the field of deep learning. Dosovitskiy et al. [36] were the first to apply a Transformer to visual tasks, proposing the Vision Transformer (ViT) as a benchmark model for image processing. The success of ViT highlights the broad adaptability of Transformer models, indicating that they are not limited to text data but can also be applied to various other data types. HSI, which contains rich spectral information and exhibits a sequential structure, has motivated researchers to develop various Transformer-based approaches, successfully applying them to HSI classification tasks [37], [38]. For example, Hong et al. [39] proposed a backbone network called SpectralFormer (SF), which reconsiders the sequential nature of the Transformer to extract local spectral sequence information from adjacent spectral bands. Compared to traditional ViT methods, this approach generates grouped spectral embeddings, leading to a substantial improvement in classification accuracy. The geometric constraints inherent in traditional convolutional kernels limit their ability to capture long-range interactions among features. To address this issue, Zhong et al. [40] introduced a novel spectral–spatial Transformer network (SSTN), which effectively overcomes these limitations by incorporating spatial attention and spectral correlation modules. This approach not only mitigates the restrictions imposed by convolution kernels but also significantly reduces computational costs. Furthermore, Chen et al. [41] developed a spectral query spatial Transformer (SQSFormer) that queries spatial features from neighboring pixels using spectral information of the central pixel. This design maintains essential spectral characteristics while effectively filtering out irrelevant information through a central focus-attention mechanism.

Although Transformer-based methods can effectively model long-range spectral dependencies, they often perform suboptimally in capturing local information due to the lack of inductive biases suitable for images, such as translation invariance. This limitation can negatively impact classification performance [42], [43]. Furthermore, Transformer models generally involve a large number of parameters and FLOPs [44], [45], demanding more training data and longer training times [46]. To address these issues, numerous studies have sought to combine the advantages of CNNs in local feature modeling with the strengths of Transformers in long-range dependency

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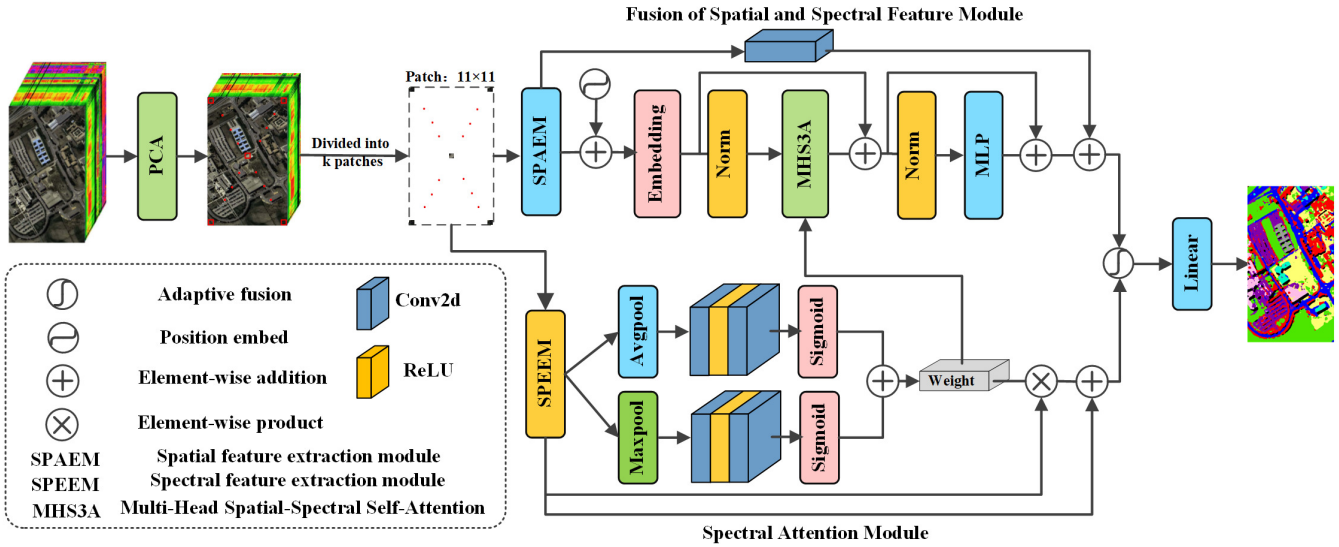



Fig. 1. Graphical illustration of the proposed S2DBFT.

modeling to optimize the extraction of spatial–spectral information. Sun et al. [47] proposed the spectral–spatial feature tokenization Transformer (SSFTT), which combines 3-D–2-D convolutional layers to extract shallow spectral and spatial features, followed by Gaussian-weighted feature tokenization for feature transformation. The transformed features are then input into a Transformer encoder for further processing. This method effectively overcomes the limitations of CNN-based classification methods in extracting deep semantic features. Zhou et al. [48] proposed a new convolutional Transformer architecture that combines CNN and Transformer models to simultaneously model the height, width, and spectral dimensions, designing a convolution-based interactive attention mechanism and extracting spatial–spectral features through parallel CNN and Transformer branches. Mei et al. [49] introduced a group-aware hierarchical Transformer (GAHT) for HSI classification, incorporating a grouped pixel embedding (GPE) module that confines MHSA to local spectral–spatial regions. Roy et al. [50] proposed a novel morphological Transformer (MorphFormer) that enhances the interaction between structural and shape information by combining spectral and spatial morphological convolution operations. In addition, Liu et al. [51] proposed a spectral–spatial enhancement and fusion network (SSEFN) specifically designed to address scenarios with limited labeled samples. By introducing a spectral–spatial enhancement strategy coupled with an adaptive decision fusion module, SSEFN not only improves the discriminability of extracted features but also effectively alleviates overfitting issues.

In recent years, while HSIC has achieved notable progress, challenges such as limited labeled data and background interference remain unresolved. To mitigate these issues, Chang and Ma [52] proposed the kernel-based constrained energy minimization (KCEM) method with bandwidth sampling (BSam), which optimizes sampling strategies and delivers strong performance in mixed-pixel classification even with fewer bands than classes. To address small-sample learning, Chang et al. [53] introduced the iterative random training sampling CNN (IRTS-CNN), incorporating a feedback loop to enhance performance under data scarcity. Ma and Chang [54] further combined KCEM with kernel-based linearly constrained minimum variance (KLCMV) to improve classification under nonlinear separability. More recently, Chang et al. [55] developed the hierarchical one-class detection (HOCD) framework, which reformulates multiclass problems into a set of one-class detection tasks. By leveraging class classification priority (CCP), HOCD effectively handles complex background scenarios, demonstrating unique advantages in background-aware HSIC (HSIC-B). These approaches not only advance theoretical modeling but also provide practical solutions for background-influenced hyperspectral scenes.

## III. Proposed Methodology

Using the University of Pavia dataset as an example, the proposed S2DBFT is described in Fig. 1. S2DBFT consists of three stages. First, the input HSI data undergo PCA for dimensionality reduction, and then, patches centered on the target pixel are fed separately into the SPAEM and SPEEM modules. Next, spatial–spectral interaction features and spectral-weighted features are obtained through the fusion of spatial and spectral feature module (FSSM) and SAM. Finally, the adaptively fused features of both branches are classified using GAP and fully connected layers, the classification result serving as the predicted label for the center pixel of the original patch. Table I presents the layer specifications and input–output dimensions of each module in the proposed S2DBFT. To facilitate the description of the dimensional changes in the feature maps, the patch size is set to $11 \times 11$, and the dimensionality of the features after PCA is set to 30.

### A. Principal Component Analysis

Let $\mathbf{X} \in \mathbb{R}^{H \times W \times S}$ be the original HSI data, where $H$ and $W$ denote the height and width of the HSI cube, respectively, and

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TABLE I
ARCHITECTURE DETAILS OF THE PROPOSED S2DBFT (TAKING THE UNIVERSITY OF PAVIA DATASET AS AN EXAMPLE)

| Stage | Module | Input size | Layer specification | Output size |
|---|---|---|---|---|
| Dimension Reduction | Encoder | (11,11,103) | PCA (dim=30) | (11,11,30) |
| Feature Extraction | SPAEM | (11,11,30) | Conv2d (inchannel=30, outchannel=30, kernelsize=3, padding=1)<br>Conv2d (inchannel=30, outchannel=30, kernelsize=5, padding=2)<br>Conv2d (inchannel=30, outchannel=30, kernelsize=7, padding=3)<br>Conv2d (inchannel=30, outchannel=30) | (11,11,30) |
| | SPEEM | (11,11,30) | Reshape, Conv1d (inchannel=30, outchannel=32)<br>Conv1d (inchannel=32, outchannel=64)<br>Conv1d (inchannel=64, outchannel=128)<br>Conv1d (inchannel=224, outchannel=256) | (121,256) |
| FSSM | Patch Embedding | (11,11,30) | Flatten, Linear (dim=256) | (121,256) |
| | Position Emdedding | (121,256) | Parameter (121,256) | (121,256) |
| | LayerNorm | (121,256) | LayerNorm (dim=256) | (121,256) |
| | MHS3A | (121,256) | Reshape, SpectralWeight<br>Conv2d (inchannel=256,outchannel=256*3) Chunk<br>MHSA (head=4,headdim=64)<br>Reshape, Linear(dim=256) | (121,256) |
| | LayerNorm | (121,256) | LayerNorm (dim=256) | (121,256) |
| | MLP | (121,256) | Linear (dim=128), GELU<br>Linear (dim=256) | (121,256) |
| SAM | Pooling | (11,11,30) | Reshape, Global average pooling, Global max pooling | (121,256) |
| | SpectralWeight | (121,256) | 2*Conv2d (inchannel=256,outchannel=16), 2*RELU<br>2*Conv2d (inchannel=16, outchannel=256), 2*Sigmoid | (121,256) |
| Pooling | GAP | (11,11,256) | Global Average Pooling | (1,256) |
| Classification | FC | (1,256) | Linear (dim=9) | (1,9) |

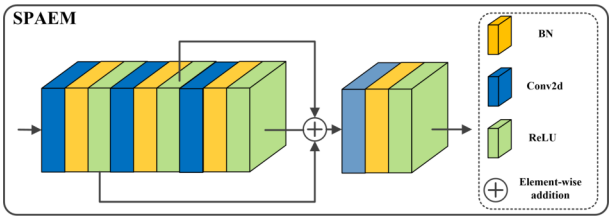



Fig. 2. Graphical illustration of the proposed SPAEM.

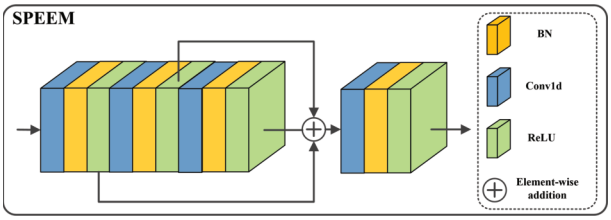



Fig. 3. Graphical illustration of the proposed SPEEM.

$S$ indicates the number of spectral bands. The HSI contains $S$ spectral bands, which provide useful spectral information while exhibiting redundancy. Therefore, we use PCA to transform $\mathbf{X}$ into $\mathbf{X}_{\text{pca}} \in \mathbb{R}^{H\times W\times c}$, reducing the number of channels to $c$. Patches are then extracted around each pixel in $\mathbf{X}_{\text{pca}}$, resulting in $\mathbf{x} \in \mathbb{R}^{h\times w\times c}$.

### B. Spectral–Spatial Feature Extraction

As shown in Fig. 2, SPAEM extracts shallow spatial features of the HSI at different scales through four 2-D convolutional layers with varying kernel sizes. Initially, the data reduced in dimension, $\mathbf{x}$, obtained through PCA, are fed into three spatial extraction blocks. Each block consists of a convolutional layer with kernel sizes of $3\times 3$, $5\times 5$, and $7\times 7$. After each convolutional layer, BN and a RELU activation function are applied. Subsequently, a 2-D convolution with a $1\times 1$ kernel size is used to effectively integrate spatial features at different scales, producing the extracted spatial features, $\mathbf{x}_{\text{spa}} \in R^{h\times w\times c}$.

For SPEEM, as shown in Fig. 3, its structure is similar to that of SPAEM and consists of three spectral extraction blocks. Each block contains a 1-D convolutional layer with a kernel size of $1\times 1$. After each convolutional layer, BN and RELU are applied. By altering the spectral dimension of $\mathbf{x}$, the shallow spectral characteristics of the HSI are extracted. Then, a 1-D convolution with a $1\times 1$ kernel size is used to integrate spectral characteristics across different dimensions, producing the extracted spectral characteristics, $\mathbf{x}_{\text{spe}} \in R^{n\times d}$, where $n = h\times w$. The corresponding formulas are as follows:

$$\mathbf{x}_{\text{spa}} = f_1\left(\text{Concat}\left(f_3\left(\mathbf{x}\right), f_5\left(f_3\left(\mathbf{x}\right)\right), f_7\left(f_5\left(f_3\left(\mathbf{x}\right)\right)\right)\right)\right) \tag{1}$$

$$f_i\left(\mathbf{x}\right) = \text{RELU}\left(\text{BN}\left(\text{Conv2d}\left(\mathbf{x}\right)\right)\right) \tag{2}$$

$$\mathbf{x}_{\text{spe}} = g_4\left(\text{Concat}\left(g_1\left(\mathbf{x}\right), g_2\left(g_1\left(\mathbf{x}\right)\right), g_3\left(g_2\left(g_1\left(\mathbf{x}\right)\right)\right)\right)\right) \tag{3}$$

$$g_i\left(\mathbf{x}\right) = \text{RELU}\left(\text{BN}\left(\text{Conv1d}\left(\mathbf{x}\right)\right)\right) \tag{4}$$

$$\text{Conv}\left(\mathbf{x}\right) = \mathbf{W}\cdot\mathbf{x} + b \tag{5}$$

$$\text{BN}\left(\mathbf{x}\right) = \frac{\mathbf{x} - \text{mean}\left(\mathbf{x}\right)}{\sqrt{\text{var}\left(\mathbf{x}\right) + \alpha}}\cdot\vec{\gamma} + \vec{\beta} \tag{6}$$

$$\text{RELU}\left(\mathbf{x}\right) = \max\left(\mathbf{x}, 0\right) \tag{7}$$

where $f_i$ represents the computation of a 2-D convolutional layer with kernel size $i$, $g_i$ represents the computation of the $i$th 1-D convolutional layer, $\mathbf{W}$ is the weight matrix, $b$ is the bias, $\alpha$ is a parameter for numerical stability, and $\vec{\gamma}$ and $\vec{\beta}$ are learnable parameter vectors.

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### C. Spectral Attention Module

To further extract spectral information from the HSI, we design the SAM, as shown in Fig. 1. The SAM consists of four 2-D convolutional layers and two RELU activation functions. Initially, the spectral features $\mathbf{x}_{\text{spe}}$, extracted by SPEEM, are separately passed through GAP and global max pooling (GMP) to compress the spatial information into a channel descriptor. This process can be expressed as follows:

$$\mathbf{x}_{\text{gap}} = \text{GAP}\left(\mathbf{x}_{\text{spe}}\right) \in R^{1\times 1\times d} \tag{8}$$

$$\mathbf{x}_{\text{gmp}} = \text{GMP}\left(\mathbf{x}_{\text{spe}}\right) \in R^{1\times 1\times d} \tag{9}$$

$$\text{GAP}\left(\mathbf{q}\right) = \frac{1}{h\times w}\sum_{i=1}^{h}\sum_{j=1}^{w}\mathbf{q}\left(i,j\right) \tag{10}$$

where $\mathbf{x}_{\text{gap}}$ and $\mathbf{x}_{\text{gmp}}$ represent the average and maximum feature maps after spatial information compression, respectively, and $\mathbf{q}$ is an input feature map.

Furthermore, for $\mathbf{x}_{\text{gap}}$, the first 2-D convolution is applied to compress $\mathbf{x}_{\text{gap}}$ into $\mathbf{x}_{\text{gap}}^{l} \in R^{1\times 1\times l}$, reducing computational complexity. Then, the RELU activation function is applied to learn the nonlinear interactions between channels in $\mathbf{x}_{\text{gap}}^{l}$. The second convolution restores the channel dimension of $\mathbf{x}_{\text{gap}}^{d} \in R^{1\times 1\times d}$, and the Sigmoid activation function is used to obtain the GAP spectral weighting $\mathbf{x}_{\text{asw}}$. For $\mathbf{x}_{\text{gmp}}$, the calculation process is similar to $\mathbf{x}_{\text{gap}}$, which is the same as the GAP output 2-D feature processing method and outputs GMP spectral weighting $\mathbf{x}_{\text{msw}}$.

Then, $\mathbf{x}_{\text{asw}}$ was combined with $\mathbf{x}_{\text{msw}}$ to obtain the spectral weight matrix $\mathbf{x}_{\text{sw}}$. $\mathbf{x}_{\text{sw}}$ is multiplied by $\mathbf{x}_{\text{spe}}$ to obtain the weighted spectral characteristics $\mathbf{x}_{\text{sw}} \in R^{N\times d}$. To prevent network degradation and improve convergence speed, $\mathbf{x}_{\text{spe}}$ is added to $\mathbf{x}_{\text{sw}}$, resulting in the final spectral feature map, $\mathbf{x}_{\text{sam}} \in R^{N\times d}$. This process can be represented as

$$\mathbf{x}_{\text{gap}}^{l} = \text{RELU}\left(\text{conv2d}\left(\mathbf{x}_{\text{gap}}\right)\right) \tag{11}$$

$$\mathbf{x}_{\text{gmp}}^{l} = \text{RELU}\left(\text{conv2d}\left(\mathbf{x}_{\text{gmp}}\right)\right) \tag{12}$$

$$\mathbf{x}_{\text{asw}} = \text{Sigmoid}\left(\text{conv2d}\left(\mathbf{x}_{\text{gap}}^{l}\right)\right) \tag{13}$$

$$\mathbf{x}_{\text{msw}} = \text{Sigmoid}\left(\text{conv2d}\left(\mathbf{x}_{\text{gmp}}^{l}\right)\right) \tag{14}$$

$$\mathbf{x}_{\text{sw}} = \mathbf{x}_{\text{asw}} + \mathbf{x}_{\text{msw}} \tag{15}$$

$$\text{Sigmoid}\left(\mathbf{x}\right) = \frac{1}{1+e^{-x}} \tag{16}$$

$$\mathbf{x}_{\text{sam}} = \mathbf{x}_{\text{sw}} * \mathbf{x}_{\text{spe}} + \mathbf{x}_{\text{spe}}. \tag{17}$$

### D. Fusion of Spatial and Spectral Feature Module

As shown in Fig. 1, the output feature **X**spa of SPAEM is used as the input to the FSSM, allowing the spatial–spectral sequence interaction. First, the feature map **X**spa is flattened into sequence data $\mathbf{x}_{\text{spa}}^{f} \in R^{N\times c}$. The tokens are then assigned to the hidden layer $\mathbf{x}_{\text{spa}}^{f1} \in R^{N\times d}$, where $d$ represents the dimension of the hidden layer. Class labels are applied, and position embedding is used to mark the positional information of each token, facilitating the final classification task while preserving positional information. This process can be expressed as follows:

$$\mathbf{x}_{\text{spa}}^{f} = \varphi_1\left(\mathbf{x}_{\text{spq}}\right) = \left[\vec{x_{11}}, \vec{x_{12}}, \ldots, \vec{x_{1n}}\right] \tag{18}$$

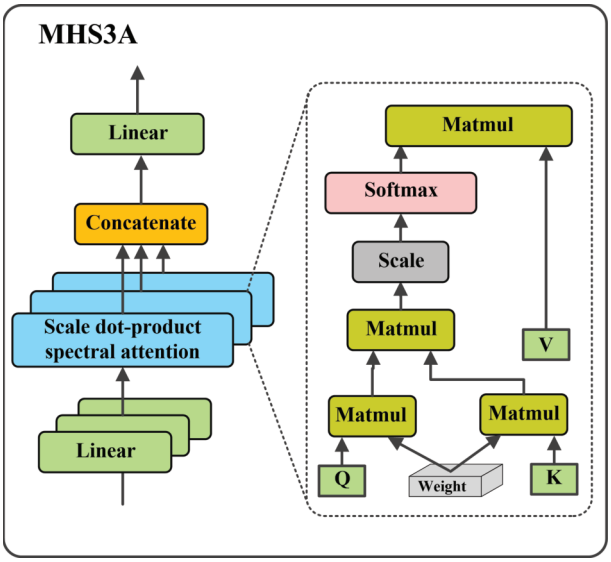



Fig. 4. Graphical illustration of the proposed MHS3A module.

$$\mathbf{x}_{\text{spa}}^{f1} = \varphi_2\left(\mathbf{x}_{\text{spa}}^{f}\right) = \left[\vec{x_{21}}, \vec{x_{22}}, \ldots, \vec{x_{2n}}\right] \tag{19}$$

$$\mathbf{x}_{f}^{p} = \left[\vec{x_{\text{cls}}}, \vec{x_{21}}, \vec{x_{22}}, \ldots, \vec{x_{2n}}\right] + \text{PE}_{\text{pos}} \tag{20}$$

where $\varphi_1$ represents the planar operation; $\varphi_2$ denotes the linear projection operation; and $\vec{x_{\text{cls}}}$ and $\text{PE}_{\text{pos}}$ refer to the classification labels and the positional information that can be learned, respectively.

Furthermore, the MHSA in ViT lacks spectral awareness, which limits its development in HSI classification. To address this issue, we designed the MHS3A module. By iteratively applying MHS3A and multilayer perceptron (MLP), the sequential features are allowed to fully interact, effectively computing global correlations. This process can be expressed as follows:

$$\mathbf{Y}_m = \text{MHS3A}\left(\text{LN}\left(\mathbf{x}_{\text{spa}}, \mathbf{x}_{\text{sw}}\right)\right) + \text{LN}\left(\mathbf{x}_{\text{spa}}, \mathbf{x}_{\text{sw}}\right) \tag{21}$$

$$\mathbf{Z}_m = \text{MLP}\left(\text{LN}\left(\mathbf{Y}_m\right)\right) + \text{LN}\left(\mathbf{Y}_m\right) \tag{22}$$

where $\mathbf{Y}_m$ and $\mathbf{Z}_m$ represent the output feature maps of MHS3A and the MLP, respectively, and LN refers to the layer normalization computation. Next, we will provide a detailed description of the modules involved in this process.

*1) Layer Normalization:* LN normalizes each channel of every sample, which stabilizes the model training, accelerates convergence, and effectively prevents gradient explosion. This process can be expressed as follows:

$$\text{LN}\left(\mathbf{x}\right) = \frac{\mathbf{x}-\boldsymbol{\mu}}{\boldsymbol{\lambda}}\cdot\varrho + \tau \tag{23}$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\lambda}$ are the mean and standard deviation of $\mathbf{x}$, respectively, and $\varrho$ and $\tau$ are learnable affine transform parameters.

*2) Multihead Spectral–Spatial Self-Attention:* As shown in Fig. 4, the **Q**, **K**, and **V** feature maps after LN are recalibrated by the spectral weights generated in SAM, which are then input into the MHSA for computation. This process compensates for the lack of spectral awareness in MHSA and enables spatial–spectral interactive feature extraction. The computation process of MHS3A can be expressed as follows:

$$\text{LN}\left(\mathbf{x}_{\text{spa}}\right) \longrightarrow \mathbf{Q}, \mathbf{K}, \mathbf{V} \tag{24}$$

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TABLE II
TRAIN AND TEST SAMPLES IN THE UNIVERSITY OF PAVIA DATASET

| ID | Land Cover Class | Train | Test |
|---|---|---|---|
| C01 | Asphalt | 33 | 6589 |
| C02 | Meadows | 93 | 18556 |
| C03 | Gravel | 10 | 2089 |
| C04 | Trees | 15 | 3049 |
| C05 | Painted metal sheets | 7 | 1338 |
| C06 | Bare Soil | 25 | 5004 |
| C07 | Bitumen | 7 | 1323 |
| C08 | Self-Blocking Bricks | 18 | 3664 |
| C09 | Shadows | 5 | 942 |
| - | Total | 213 | 42563 |

$$\mathbf{Q}_s = Q * \mathbf{x}_{\text{sw}}, K_s = K * \mathbf{x}_{\text{sw}} \tag{25}$$

$$S3A = \text{Attention}(\mathbf{Q}_s, \mathbf{K}_s, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}_s \mathbf{K}_s^{\text{T}}}{d_{sk}}\right)\mathbf{V} \tag{26}$$

$$\text{MHS3A} = \text{Concat}(S3A_1, S3A_2, \ldots, S3A_h)\,\mathbf{W} \tag{27}$$

where $\mathbf{Q}_s$, $\mathbf{K}_s$, and $\mathbf{V}$ are learnable weight matrices; $\mathbf{x}_{\text{spq}^{\text{sw}}}$ represents the recalibrated features; $d_{sk}$ is the dimensionality of $\mathbf{K}_s$; $h$ denotes the number of heads; and $\mathbf{W}$ refers to the parameter matrix.

*3) Multilayer Perceptron:* The MLP consists of a linear projection layer and a GELU activation function, which are used to capture the long-term dependencies of the tokens in the sequence. The specific formula is as follows:

$$\mathbf{x}_{\text{fssm}} = \text{MLP}(z_m) = \text{Linear}(\text{GELU}(\text{Linear}(z_m))) \tag{28}$$

$$\text{GELU}(\mathbf{y}) = 0.5\mathbf{y}\left(1 + \text{Tanh}\left(\sqrt{\frac{2}{\pi}}\left(\mathbf{y} + 0.044715\mathbf{y}^3\right)\right)\right) \tag{29}$$

where $\mathbf{x}_{\text{fssm}}$ is the ouput of FSSM and $\mathbf{y}$ is an input feature map.

### E. Adaptive Fusion and Classification

First, the global spatial–spectral weighted interaction features obtained from FSSM are fused with the spatial features output by SPAEM through residual fusion. Simultaneously, the local spectral weighted features derived from SAM are integrated with the spectral features extracted by SPEEM using the same fusion strategy. This residual fusion mechanism effectively alleviates the decline in model accuracy caused by network degradation. The detailed process is illustrated as follows:

$$\mathbf{x}^1_{\text{fusion}} = \mathbf{x}_{\text{fssm}} + \text{conv2d}\left(\mathbf{x}_{\text{spa}}\right) \tag{30}$$

$$\mathbf{x}^2_{\text{fusion}} = \mathbf{x}_{\text{sam}} + \mathbf{x}_{\text{spe}} \tag{31}$$

where $\mathbf{x}^1_{\text{fusion}}$ represents the residual fusion feature map of FSSM and SPAEM, and $\mathbf{x}^2_{\text{fusion}}$ denotes the residual fusion feature map of SAM and SPEEM. The Conv2d operation is employed to integrate the features output by SPAEM and FSSM into the same vector space, ensuring compatibility for the subsequent residual connection process.

Then, $\mathbf{x}^1_{\text{fusion}}$ and $\mathbf{x}^2_{\text{fusion}}$ are adaptively fused to more effectively capture the discriminative features between different classes, thereby further enhancing classification performance. The specific process is as follows:

$$\mathbf{x}_{\text{fusion}} = \alpha * \mathbf{x}^1_{\text{fusion}} + (1 - \alpha)\,\mathbf{x}^2_{\text{fusion}} \tag{32}$$

where $\alpha$ is the adaptive normalized weight. In the model training process, the assigned weights of the spectrum and space are adjusted to achieve the minimum loss value and obtain the optimal weight value. The initial value of $\alpha$ is set to 0.5, and the optimal weights are updated each time the model is run.

Finally, the training loss is computed using the cross-entropy loss function, and the model weights are updated. The specific process is as follows:

$$L_{\text{CE}} = \frac{-\sum_{i=1}^{N} l_i \log\left(l_i^{\epsilon}\right)}{N} \tag{33}$$

where $N$ is the total number of training samples and $l^i$ and $l_i^{\epsilon}$ are the true label and the predicted class probabilities of the $i$th sample, respectively.

TABLE III
TRAIN AND TEST SAMPLES IN THE SALINAS DATASET

| ID | Land Cover Class | Train | Test |
|---|---|---|---|
| C01 | Brocoli green weeds 1 | 10 | 1999 |
| C02 | Brocoli green weeds 2 | 19 | 18556 |
| C03 | Fallow | 10 | 1996 |
| C04 | Fallow rough plow | 7 | 1387 |
| C05 | Fallow smooth | 13 | 2665 |
| C06 | Stubble | 20 | 3939 |
| C07 | Celery | 18 | 3561 |
| C08 | Grapes untrained | 56 | 11215 |
| C09 | Soil vinyard develop | 31 | 6172 |
| C10 | Corn senesced green weeds | 16 | 3262 |
| C11 | Lettuce romaine 4wk | 5 | 1063 |
| C12 | Lettuce romaine 5wk | 10 | 1917 |
| C13 | Lettuce romaine 6wk | 5 | 911 |
| C14 | Lettuce romaine 7wk | 5 | 1065 |
| C15 | Vinyard untrained | 36 | 7232 |
| C16 | Vinyard vertical trelis | 9 | 1798 |
| - | Total | 270 | 53859 |

## IV. EXPERIMENTAL RESULTS

In this section, we validate the effectiveness of the proposed method through a series of comprehensive experiments conducted on four publicly available datasets: University of Pavia, Salinas, WHU-Hi-LongKou, and Botswana. The number of labeled samples corresponding to different categories within each dataset is presented in Tables II–V. The false-color composite images and the corresponding ground truth (GT) maps for four datasets are illustrated in Figs. 5–8. Initially, a comprehensive description of the four datasets employed in the experiments is provided, detailing their characteristics and relevance to the study. Subsequently, we outline the various experimental configurations, followed by an in-depth analysis of the parameter settings used in the study. Next, we present the classification results obtained from the proposed method as well as those derived from the comparison methods, accompanied by a thorough analysis of the outcomes. Finally, we conduct an ablation study and explore the gradient-based analysis of the training samples, shedding light on their impact on model performance.

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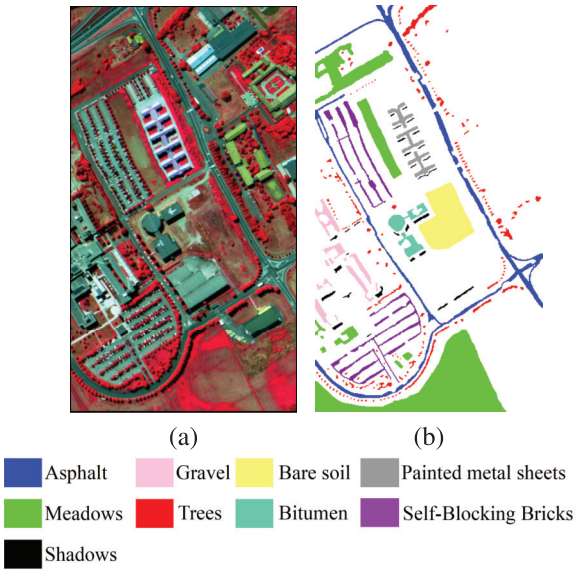



Fig. 5. University of Pavia. (a) False-color composite image (R: 102, G: 56, and B: 31). (b) GT map.

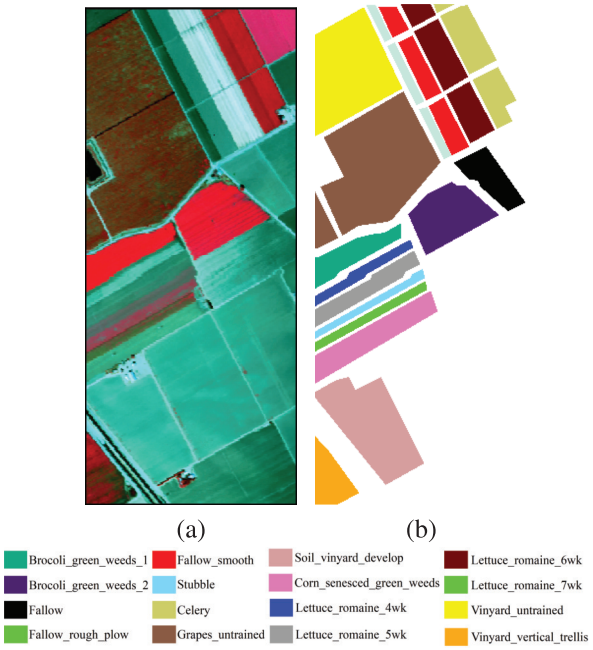



Fig. 6. Salinas. (a) False-color composite image (R: 57, G: 27, and B: 17). (b) GT map.

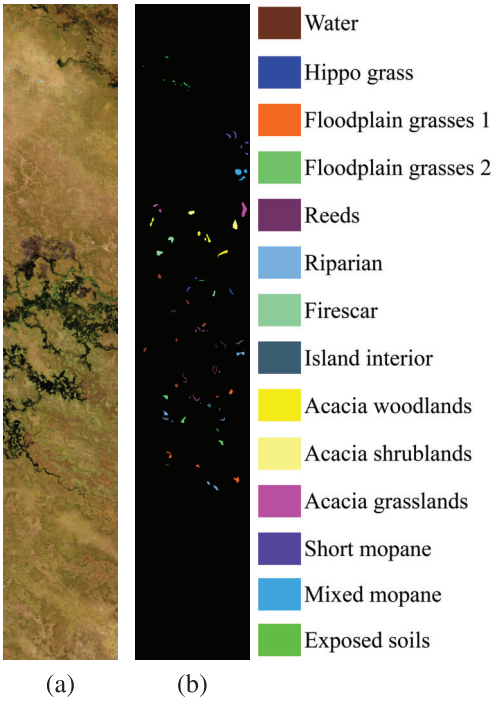



Fig. 7. Botswana. (a) False-color composite image (R: 75, G: 33, and B: 15). (b) GT map.

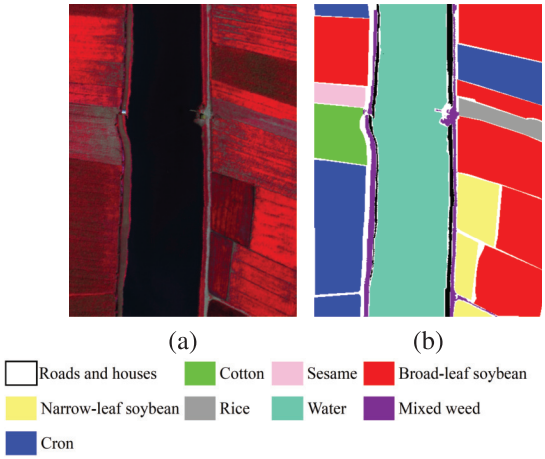



Fig. 8. WU-HI-LongKou. (a) False-color composite image (R: 180, G: 126, and B: 72). (b) GT map.

## A. Datasets' Description

*1) University of Pavia:* The University of Pavia dataset, collected by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor during an aerial survey over Pavia, Northern Italy, in 2001, features a spatial resolution of 1.3 m and dimensions of 610 × 340 pixels. It originally contained 115 spectral bands ranging from 0.43 to 0.86 $\mu$m, but after removing 12 noise bands, 103 bands were used in experiments. The dataset encompasses nine distinct land-cover categories, with pseudocolor and GT maps illustrated in Fig. 5.

*2) Salinas:* The Salinas dataset was acquired in the Salinas Valley of California, USA, using advanced hyperspectral imaging techniques. It has a spatial resolution of 3.7 m and comprises 512 × 217 pixels distributed over 204 spectral bands. The dataset captures a diverse range of features and includes 16 distinct land-cover categories. These categories highlight various agricultural and natural elements of the region. Its pseudocolor representation and GT map are provided in Fig. 6.

*3) Botswana:* The Botswana dataset, captured by NASA's EO-1 satellite over the Okavango Delta, featured a 7.7-km-long data strip acquired with the Hyperion sensor. It has a spatial resolution of 30 m/pixel and initially included 242 spectral bands spanning the 400–2500-nm range in 10-nm intervals. After excluding absorbent bands, 145 spectral bands remain. The dataset is categorized into 14 distinct classes. Its pseudocolor representation and GT map are provided in Fig. 7.

*4) WHU-HI-LongKou:* The WHU-HI-LongKou dataset was collected on July 17, 2018, in Longkou, Hubei, China, using a Headwall Nano-Hyperspec imaging sensor with an 8-mm focal length. The imagery, measuring 550 × 400 pixels,

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TABLE IV
TRAIN AND TEST SAMPLES IN THE BOTSWANA DATASET

| ID | Land Cover Class | Train | Test |
|---|---|---|---|
| C01 | Water | 6 | 264 |
| C02 | Hippo grass | 2 | 99 |
| C03 | Floodplain grasses 1 | 5 | 246 |
| C04 | Floodplain grasses 2 | 4 | 211 |
| C05 | Reeds | 5 | 2664 |
| C06 | Riparian | 5 | 264 |
| C07 | Firescar | 5 | 254 |
| C08 | Island interior | 4 | 199 |
| C09 | Acacia woodlands | 6 | 308 |
| C10 | Acacia shrublands | 5 | 243 |
| C11 | Acacia grasslands | 6 | 299 |
| C12 | Short mopane | 4 | 177 |
| C13 | Mixed mopane | 5 | 263 |
| C14 | Exposed soils | 2 | 93 |
| - | Total | 64 | 3184 |

TABLE V
TRAIN AND TEST SAMPLES IN THE WHU-HI-LONGKOU DATASET

| ID | Land Cover Class | Train | Test |
|---|---|---|---|
| C01 | Corn | 69 | 34442 |
| C02 | Cotton | 17 | 8357 |
| C03 | Sesame | 6 | 3025 |
| C04 | Broad-leaf soybean | 126 | 63086 |
| C05 | Narrow-leaf soybean | 8 | 4143 |
| C06 | Rice | 24 | 11830 |
| C07 | Water | 134 | 66922 |
| C08 | Roads and houses | 14 | 7110 |
| C09 | Mixed weed | 11 | 5218 |
| - | Total | 409 | 204133 |

comprises 270 spectral bands spanning wavelengths from 0.400 to 1.0 $\mu$m. The study area features a straightforward agricultural landscape with nine land-cover classes, including six crop species. Its pseudocolor representation and GT map are provided in Fig. 8.

## B. Experimental Settings

*1) Hyperparameters:* In our approach, the Adam optimizer is employed to update the network parameters, with a learning rate set at $5e^{-4}$. The training process halts when the training loss stabilizes across epochs. For the experiments conducted on the four datasets, the number of training epochs is consistently set to 50, ensuring stability in the training loss under this configuration. For all datasets, the batch size is set to 64. Additionally, in the parameter sensitivity analysis, hyperparameters, including the number of principal components (PCs), patch size, learning rate, and the number of heads in MHS3A of the FSSM module were analyzed. All comparative methods utilized the original code provided by the authors and were re-executed on the same hardware configuration to ensure experimental consistency.

*2) Training Samples:* To assess the model's classification performance under limited sample conditions, we partitioned the number of samples used in the hyperparameter tuning experiments, comparative experiments, and ablation studies. Specifically, for the University of Pavia and Salinas datasets, 0.5% of the samples were selected for training; for the Botswana dataset, 2% was chosen; and for the WHU-Hi-LongKou dataset, 0.2% was also used. All training samples were randomly selected to ensure the stability of the algorithm, with the remaining samples used exclusively for testing purposes.

*3) Performance Comparison:* The proposed method is compared with other state-of-the-art deep learning approaches, including IRTS-3DCNN [53], HybridSN [34], double-branch dual-attention (DBDA) [35], SF [39], SSFTT [47], GAHT [49], SSTN [40], MorphFormer [50], spectral–spatial masked Transformer (SS-MTr) [56], SQSFormer [41], and SSEFN [51]. The parameter settings for all comparison methods are consistent with those reported in the corresponding articles.

*4) Running Platforms and Metrics:* To ensure a fair comparison, both the proposed method and all comparative methods are based on the PyTorch 2.0.0 and cuDNN 11.8 framework. Training and testing are carried out on a system equipped with an Intel[1] Core[2] i7-14700 processor and an NVIDIA GeForce RTX 4060 GPU (8 GB). For quantitative evaluation of the classification performance, four commonly used metrics are employed. The per-class accuracy represents the ratio of correctly classified samples to the total number of samples in each class. Overall accuracy (OA) is the ratio of all correctly classified samples to the total number of samples. The average accuracy (AA) is the arithmetic mean of the per-class accuracies. The kappa coefficient ($\kappa$) measures the consistency between the model's classification results and the GT labels. PR is the ratio of correctly predicted positive samples to the total number of predicted positive samples for a given class. For all these metrics, higher values indicate better performance of the method.

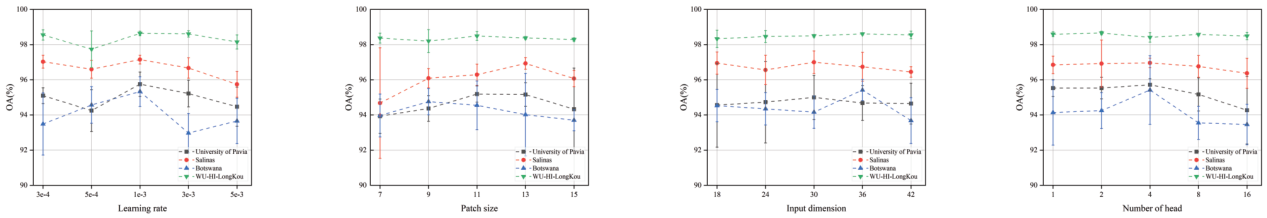
## C. Parameter Sensitive Analysis

*1) Learning Rate:* The learning rate is a crucial hyperparameter in the model training process, directly influencing the convergence speed of the model. In order to determine the optimal learning rate for each dataset, we conducted a series of experiments using a set of candidate values to [$3e^{-4}$, $5e^{-4}$, $1e^{-3}$, $3e^{-3}$, $5e^{-3}$]. Experiments were carried out on the four datasets mentioned, and the OA values were recorded for each corresponding learning rate. As depicted in Fig. 9(a), for all four datasets, the highest OA was achieved when the learning rate was set to 0.001. Moreover, both increasing and decreasing the learning rate resulted in a noticeable reduction in OA. Consequently, in the subsequent experiments conducted using the proposed method, the learning rate was consistently maintained at 0.001.

*2) Patch Size:* To investigate the impact of patch size on classification performance, we maintained all other parameters constant and considered candidate patch sizes of [7, 9, 11, 13, 15]. Fig. 9(b) illustrates the model's OA across four datasets under different patch sizes. For the University of Pavia, Salinas, and Botswana datasets, the OA values exhibit an initial increase followed by a subsequent decrease as the patch size increases within the candidate range, while for the WHU-Hi-LongKou dataset, the OA values show some fluctuations but remain generally stable across the candidate

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(a) (b) (c) (d)

Fig. 9. Evolution of OA as a function of (a) learning rate, (b) patch size, (c) input dimension, and (d) number of heads.

TABLE VI
ABLATION STUDY OF MODULES OF S2DBFT

| SPAEM | SPEEM | FSSM | SAM | University of Pavia | Salinas | Botswana | WU-HI-LongKou |
|---|---|---|---|---|---|---|---|
| ✓ | ✗ | ✗ | ✗ | 52.78±0.84 | 75.91±3.44 | 33.05±5.84 | 90.47±1.37 |
| ✗ | ✓ | ✗ | ✗ | 72.58±2.71 | 85.02±0.71 | 51.96±6.43 | 92.58±0.96 |
| ✓ | ✓ | ✗ | ✗ | 82.24±2.44 | 94.45±0.64 | 62.68±3.01 | 97.22±0.49 |
| ✓ | ✗ | ✓ | ✗ | 94.71±1.02 | 96.46±0.48 | 94.35±1.72 | 98.44±0.16 |
| ✗ | ✓ | ✗ | ✓ | 75.09±2.72 | 85.29±2.62 | 62.09±2.39 | 93.23±0.76 |
| ✓ | ✓ | ✓ | ✗ | 94.42±1.38 | 96.59±0.62 | 94.94±0.87 | 98.58±0.22 |
| ✓ | ✓ | ✗ | ✓ | 86.55±1.56 | 95.22±0.75 | 68.31±1.73 | 97.27±0.47 |
| ✓ | ✗ | ✓ | ✓ | 95.11±1.31 | 96.67±0.83 | 94.57±0.84 | 98.52±0.11 |
| ✗ | ✓ | ✓ | ✓ | 95.15±1.34 | 96.88±0.33 | 94.67±0.73 | 98.51±0.08 |
| ✓ | ✓ | ✓ | ✓ | **95.72±0.34** | **96.96±0.27** | **95.33±0.84** | **98.64±0.13** |

range. When the patch size is set to $11 \times 11$, the OA reaches its peak for University of Pavia and WHU-Hi-LongKou datasets, achieving 95.19%, and 98.49%, respectively. For the Salinas and Botswana datasets, when the patch size is set to $11 \times 11$, the performance of OA is optimal, reaching 96.93% and 94.76%. Therefore, based on the above experimental results, we select a patch size of 11 for all four datasets.

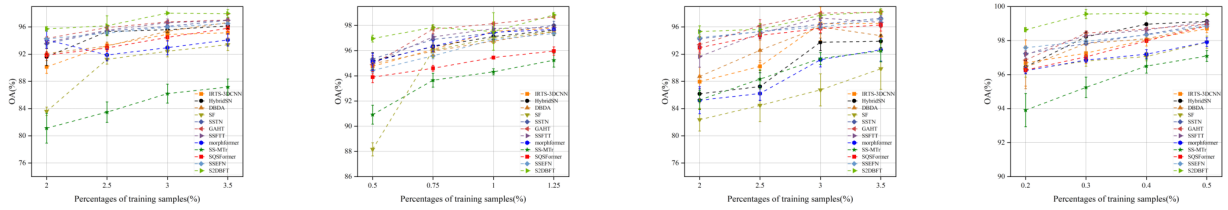
*3) PCA Dimensionality Reduction:* To identify the optimal dimensionality reduction using PCA, we examined how varying the number of PCs [18, 24, 30, 36, 42] influenced classification performance, with all other parameters held constant. We evaluated OA under these dimensions. As illustrated in Fig. 9(c), the OA changes according to the number of PCs. For the Salinas and Botswana datasets, the effect of PCA dimensionality reduction on OA showed fluctuations. Among the selected spectral dimensions, as the number of bands increased, OA varied for both datasets, peaking at a dimensionality of 36 (97.00% and 95.42%). In contrast, the University of Pavia dataset demonstrated a more pronounced response to PCA-based reduction, with a marked improvement in OA within the tested range. Specifically, OA initially increased before declining, with an optimal accuracy of 95.00% at 36 dimensions. For the WHU-Hi-LongKou dataset, the effect of PCA-based dimensionality reduction was less significant, as OA remained fairly stable across the candidate values, although the accuracy at 30 dimensions (98.61%) was marginally higher than at other settings. Considering both classification performance and computational complexity, we determined that 30 spectral dimensions offered a suitable balance and consequently selected this configuration for subsequent experiments.

*4) Number of Heads in MHS3A:* In order to evaluate the impact of different head numbers in MHS3A on model performance, we set a set of candidate values for the number of heads ($H$), including [1, 2, 4, 8, 16]. As illustrated in Fig. 9(d), it can be observed that for the University of Pavia, Salinas, and Botswana datasets, the accuracy initially increases and subsequently decreases with the growing number of heads, reaching peak performance when $H = 4$. For the WHU-Hi-LongKou dataset, despite minor fluctuations across different values, the variation remains relatively modest within the candidate range, achieving optimal accuracy at $H = 2$. Consequently, based on comprehensive evaluations across these four benchmark datasets, $H = 4$ is selected as the optimal configuration to ensure that the proposed model delivers superior performance on most datasets.

## D. Ablation Study

*1) Effectiveness of SPAEM and SPEEM:* In the proposed S2DBFT, the SPAEM and SPEEM effectively extract the shallow spatial–spectral features of HSI. To validate their effectiveness, we conducted experiments by removing either SPAEM or SPEEM. As shown in Table VI, when neither the FSSM nor the SAM was used, the individual use of SPAEM and SPEEM performed poorly on both the University of Pavia and Botswana datasets. However, when both SPAEM and SPEEM were used together, the OAs of the University of Pavia increased by 29.46% and 9.66%, respectively, compared to the individual use of SPAEM and SPEEM. The OAs for Salinas increased by 18.46% and 9.43%; for Botswana, the OAs improved by 29.63% and 10.72%; and for WU-HI-LongKou, the OAs increased by 6.78% and 4.64%. With the FSSM and SAM already in use, the application of the SPEEM further improved the OA performance on all four datasets, with the most significant improvement observed on the University of Pavia dataset (an increase of 0.43%). Additionally, improvements were observed on Salinas, Botswana, and WU-HI-LongKou, with increases of 0.26%, 0.21%, and 0.08%, respectively. After adding the SPAEM, the OAs for

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(a) (b) (c) (d)

Fig. 10. Evolution of OA as a function of training samples per class. (a) University of Pavia. (b) Salinas. (c) Botswana. (d) WU-HI-LongKou.

TABLE VII
ABLATION STUDY OF MHS3A MODULE IN FOUR DATASETS

| Datasets | FSSM | | OA (%) | AA (%) | $\kappa \times 100$ |
|---|---|---|---|---|---|
| | MHSA | MHS3A | | | |
| University of Pavia (PU) | ✓ | ✗ | 95.43±0.39 | 92.22±0.64 | 93.92±0.51 |
| | ✗ | ✓ | **95.72±0.34** | 93.11±0.70 | **94.32±0.45** |
| Salinas (SA) | ✓ | ✗ | 95.92±0.44 | 96.64±0.38 | 95.57±0.49 |
| | ✗ | ✓ | **96.96±0.27** | **97.63±0.34** | **96.61±0.30** |
| Botswana (BT) | ✓ | ✗ | 95.09±1.31 | 94.69±1.28 | 94.68±1.42 |
| | ✗ | ✓ | **95.33±0.84** | **94.90±0.80** | **94.94±0.91** |
| WU-HI-LongKou (LK) | ✓ | ✗ | 97.70±1.27 | 94.13±2.82 | 96.98±1.67 |
| | ✗ | ✓ | **98.64±0.13** | 95.76±0.44 | **98.21±0.17** |

University of Pavia, Salinas, Botswana, and WU-HI-LongKou increased by 0.57%, 0.08%, 0.66%, and 0.13%, respectively.

*2) Effectiveness of MHS3A:* To further validate the effectiveness of MHS3A, we replaced it with MHSA. As indicated in Table VII, compared to using MHSA, S2DBFT showed a significant increase in OA, AA, and Kappa across all four datasets, with the most notable improvement observed in AA. To be specific, the AAs for University of Pavia, Salinas, Botswana, and WU-HI-LongKou increased by 0.89%, 0.99%, 0.21%, and 1.63%, respectively.

### E. Assessing Results With Varied Training Samples

To further evaluate the scalability and generalization ability of the proposed method in this section, we investigate the classification performance of the S2DBFT method alongside other competing approaches under varying training sample proportions. The OA metric is adopted as the evaluation criterion; however, considering the varying classification performances across different datasets, we divide the four datasets according to the following sample ratios: 0.50%, 0.75%, 1.00%, and 1.25% for University of Pavia and Salinas; 2.00%, 2.50%, 3.00%, and 3.50% for Botswana; and 0.20%, 0.30%, 0.40%, and 0.50% for WHU-Hi-LongKou. Fig. 10 presents the corresponding experimental results on different HSI classification datasets. Under all training sample conditions for University of Pavia and WHU-Hi-LongKou, the S2DBFT method consistently achieved the highest accuracy. Under the second and third gradient sample conditions on the Botswana dataset, the GAHT [49] method slightly outperforms our approach. This can be attributed to the spectral mixing phenomenon at land-cover boundaries in the Botswana dataset, which requires more refined spectral–spatial decoupling. Our SAM module, when trained with insufficient samples, may excessively suppress certain critical spectral bands, leading to suboptimal handling of mixed pixels. On the Salinas dataset, our method exhibits notable performance fluctuations under the third gradient sample condition. Given that the Salinas dataset consists of large, homogeneous regions with spectrally similar land-cover classes, the model may over-rely on spectral information while underutilizing spatial context. The FSSM module, which depends on robust spectral–spatial interaction, may demonstrate reduced fusion stability when training samples are limited, indicating room for further optimization in such scenarios. Nevertheless, as the number of training samples increases, the proposed S2DBFT method consistently achieves superior accuracy across all four datasets, demonstrating its strong scalability and generalization capability.

### F. Classification Results

To evaluate the effectiveness of the proposed S2DBFT, we conducted comparative experiments against CNN- and Transformer-based methods. These include nine advanced HSI classification algorithms: IRTS-3DCNN [53], HybridSN [34], DBDA [35], SF [39], SSFTT [47], GAHT [49], SSTN [40], MorphFormer [50], SS-MTr [56], SQSFormer [41], and SSEFN [51]. Four publicly available datasets—University of Pavia, Salinas, Botswana, and WHU-Hi-LongKou—were used as research objects for a series of experiments. To ensure fairness, all experiments were conducted under identical environmental conditions and with the hyperparameters specified in the original articles. The classification results for each method on the datasets are provided in Tables V–VIII, with the corresponding classification maps shown in Figs. 6–9. The classification results include overall accuracy (OA, %), average accuracy (AA, %), Kappa coefficient ($\kappa$, %), and precision (PR, %), along with the standard deviation after five independent experiments, with the best results highlighted in bold. The following is a detailed analysis of the experimental results.

From the overall results, it is evident that S2DBFT outperforms all other methods across all four datasets, with its OA, AA, Kappa, and PR values generally surpassing those of other methods in the majority of cases, particularly in terms of classification accuracy in complex scenes. The Kappa coefficient of S2DBFT is significantly higher than that of the competing algorithms, suggesting that its classification results exhibit a higher level of agreement with the true values, reflecting its high reliability and stability in HSIC tasks. Compared to other methods, S2DBFT is more efficient in capturing deep features of spectral–spatial information, further enhancing the performance of the model.

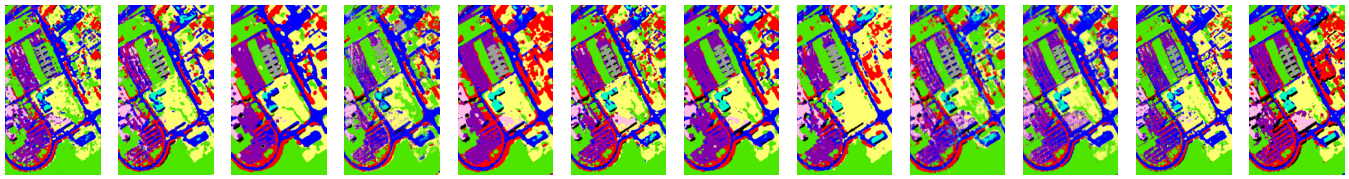
*1) University of Pavia:* On the University of Pavia dataset, the classification accuracy tables and maps for each method

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TABLE VIII

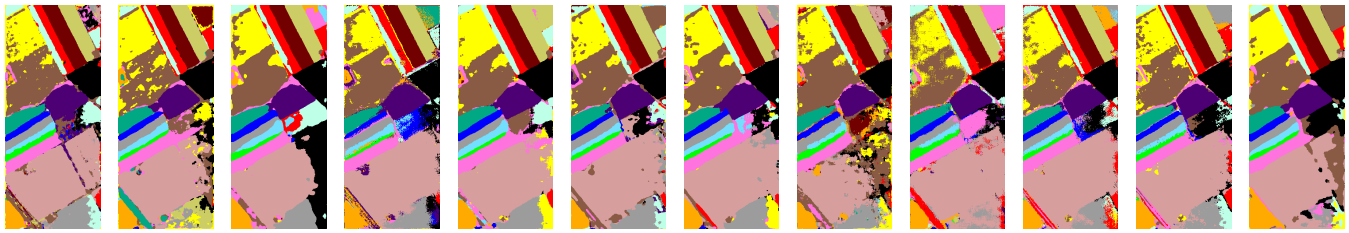
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS IN THE UNIVERSITY OF PAVIA (0.5% FOR TRAINING)

| Class | IRTS-3DCNN [53] | HybridSN [34] | DBDA [35] | SF [39] | GAHT [49] | SSFTT [47] | SSTN [40] | morphformer [50] | SS-MTr [56] | SQSFormer [41] | SSEFN [51] | **S2DBFT (Ours)** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 87.77±3.92 | 91.32±4.21 | 90.84±0.71 | 87.09±2.15 | 85.00±4.89 | 87.32±1.42 | 83.79±2.44 | **93.88±2.23** | 81.96±2.78 | 87.92±2.12 | 92.48±0.50 | 92.35±2.22 |
| 2 | 99.71±0.19 | 98.86±0.82 | 99.61±0.10 | 96.51±1.32 | 99.92±0.10 | 99.95±0.07 | 99.95±0.08 | 98.69±1.09 | 96.81±1.09 | 99.47±0.16 | 99.67±0.08 | **99.96±0.28** |
| 3 | 68.43±4.23 | **83.32±7.22** | 63.61±0.45 | 62.84±5.01 | 71.39±4.33 | 74.48±5.29 | 71.07±5.35 | 67.71±12.54 | 50.46±9.18 | 67.48±5.43 | 68.01±1.65 | 74.81±9.56 |
| 4 | 83.04±3.34 | 80.00±5.08 | 84.46±2.97 | 84.00±3.04 | 90.10±1.62 | 84.26±2.05 | 84.20±1.78 | **93.24±2.14** | 71.01±6.94 | 87.70±1.12 | 88.33±0.50 | 88.76±7.04 |
| 5 | 99.15±0.82 | 99.63±0.39 | 99.99±0.03 | 99.80±0.28 | 99.97±0.06 | 99.90±0.13 | 99.84±0.18 | 98.29±1.39 | 99.52±0.45 | **100.00±0.00** | **100.00±0.00** | 99.48±0.41 |
| 6 | 84.88±1.87 | 94.51±6.53 | 94.76±1.19 | 61.72±5.30 | 97.55±0.87 | 97.47±1.14 | 97.28±2.00 | 94.90±4.91 | 64.25±13.90 | 85.96±1.77 | 90.57±1.08 | **99.31±0.56** |
| 7 | 81.81±4.64 | 90.56±9.76 | 73.64±13.55 | 54.33±6.29 | 99.24±0.39 | 98.91±1.17 | 98.19±1.15 | 75.79±8.55 | 58.16±23.02 | 94.88±2.20 | 96.04±0.90 | **99.82±0.26** |
| 8 | 76.05±5.27 | 72.22±7.52 | 89.10±1.33 | 55.07±3.27 | **95.43±1.56** | 89.26±1.93 | 92.61±2.88 | 91.53±1.97 | 59.78±5.68 | 81.05±4.92 | 92.73±0.99 | 92.88±8.11 |
| 9 | 69.02±19.43 | 56.39±7.02 | 53.40±27.28 | 95.40±1.43 | 71.44±12.96 | 70.45±16.64 | 83.35±4.30 | 94.74±2.79 | 45.99±15.35 | 94.59±1.36 | **97.30±0.70** | 90.87±3.82 |
| OA (%) | 90.13±1.41 | 91.60±1.28 | 92.10±0.72 | 83.61±0.62 | 94.16±1.04 | 93.72±0.25 | 93.53±0.45 | 94.13±1.28 | 81.13±2.19 | 91.86±0.47 | 94.24±0.10 | **95.72±0.34** |
| AA (%) | 83.34±2.64 | 85.30±1.48 | 83.27±3.38 | 77.42±0.99 | 90.01±2.02 | 89.11±1.08 | 90.03±1.00 | 89.75±2.28 | 69.77±3.21 | 88.78±0.65 | 90.68±0.24 | **93.11±0.70** |
| $\kappa \times 100$ | 86.72±1.91 | 88.79±1.71 | 89.46±0.98 | 77.88±0.86 | 92.28±1.38 | 91.63±0.34 | 91.39±0.61 | 92.21±1.70 | 74.44±3.24 | 89.12±0.63 | 92.28±0.14 | **94.32±0.45** |
| PR (%) | 89.22±0.84 | 92.84±0.35 | 84.57±1.40 | 84.26±2.21 | 94.71±0.56 | 93.86±0.25 | 93.83±0.10 | 92.11±1.01 | 78.10±3.25 | 92.08±0.66 | 95.23±0.09 | **95.95±0.78** |



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 11. Classification maps obtained by different methods in the University of Pavia (0.5% for training). (a) IRTS-3DCNN (90.13%). (b) HybridSN (91.60%). (c) DBDA (92.10%). (d) SF (83.61%). (e) GAHT (94.16%). (f) SSFTT (93.72%). (g) SSTN (93.53%). (h) MorphFormer (94.13%). (i) SS-MTr (81.13%). (j) SQSFormer (91.86%). (k) SSEFN (94.24%). (l) S2DBFT (95.72%).



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 12. Classification maps obtained by different methods in Salinas (0.5% for training). (a) IRTS-3DCNN (94.76%). (b) HybridSN (95.19%). (c) DBDA (94.96%). (d) SF (88.17%). (e) GAHT (94.90%). (f) SSFTT (95.01%). (g) SSTN (95.33%). (h) MorphFormer (95.11%). (i) SS-MTr (90.92%). (j) SQSFormer (93.90%). (k) SSEFN (94.43%). (l) S2DBFT (96.96%).

TABLE IX

CLASSIFICATION ACCURACIES OF DIFFERENT METHODS IN SALINAS (0.5% FOR TRAINING)

| Class | IRTS-3DCNN [53] | HybridSN [34] | DBDA [35] | SF [39] | GAHT [49] | SSFTT [47] | SSTN [40] | morphformer [50] | SS-MTr [56] | SQSFormer [41] | SSEFN [51] | **S2DBFT (Ours)** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 98.29±1.13 | 96.11±2.39 | 95.45±2.53 | 93.14±1.98 | **99.56±0.89** | 96.02±1.60 | 95.19±1.47 | 97.17±5.17 | 97.11±1.96 | 96.01±1.25 | 96.90±0.64 | 97.64±1.92 |
| 2 | 99.88±0.11 | 99.97±0.05 | **100.00±0.00** | 98.75±0.62 | 99.72±0.60 | **100.00±0.00** | **100.00±0.00** | 99.17±2.09 | 99.86±0.14 | 99.99±0.01 | 99.96±0.01 | 99.98±0.03 |
| 3 | 99.88±0.13 | 99.77±0.26 | 99.98±0.04 | 93.57±2.74 | 97.16±2.24 | 99.90±0.12 | **100.00±0.00** | 93.44±7.90 | 98.48±1.47 | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** |
| 4 | 92.04±3.55 | 94.51±3.09 | 97.59±1.82 | 90.39±1.48 | 98.32±1.40 | 95.43±3.66 | 97.09±1.07 | 94.05±3.08 | 94.88±2.15 | 90.63±2.03 | 89.17±1.03 | **99.39±0.49** |
| 5 | 91.8±1.61 | 96.45±0.45 | 95.25±1.12 | 81.20±1.61 | 97.10±2.32 | 91.27±2.67 | 94.07±1.15 | 93.80±4.13 | 82.68±15.35 | **97.85±0.62** | 97.12±0.13 | 95.08±1.53 |
| 6 | 100.00±0.00 | 99.96±0.06 | **100.00±0.00** | 99.90±0.17 | 99.99±0.03 | **100.00±0.00** | **100.00±0.00** | 98.44±2.47 | 99.31±0.61 | 99.99±0.01 | **100.00±0.00** | 99.84±0.10 |
| 7 | 99.90±0.07 | 99.81±0.23 | 99.97±0.06 | 98.69±0.39 | 99.73±0.30 | 99.94±0.04 | 99.89±0.12 | 98.90±1.03 | 99.51±0.39 | 99.97±0.07 | **99.98±0.01** | 99.68±0.20 |
| 8 | 91.04±1.77 | 89.36±2.42 | 91.28±0.61 | 80.56±2.53 | 88.86±3.46 | 92.67±0.52 | 89.84±1.99 | **93.18±2.35** | 78.90±7.07 | 86.77±0.86 | 87.86±0.77 | 92.01±1.29 |
| 9 | 99.93±0.05 | 99.99±0.01 | 99.97±0.05 | 94.67±0.95 | 98.62±1.45 | 99.98±0.04 | **100.00±0.00** | 99.58±0.50 | 99.89±0.10 | **100.00±0.00** | **100.00±0.00** | 99.90±0.19 |
| 10 | 98.77±0.42 | 98.44±0.42 | 99.10±0.31 | 88.43±1.36 | 98.46±1.33 | 98.05±0.57 | **99.22±0.32** | 96.49±3.92 | 92.33±6.74 | 98.77±0.14 | 97.73±0.18 | 99.02±0.28 |
| 11 | 99.21±0.71 | 99.21±0.31 | 99.68±0.35 | 75.72±5.62 | 91.43±6.62 | 97.31±1.18 | **100.00±0.00** | 95.63±5.63 | 95.45±8.69 | 99.77±0.33 | **100.00±0.00** | 99.98±0.04 |
| 12 | 98.08±1.17 | 97.79±1.62 | 98.84±0.72 | 97.37±1.17 | 99.37±0.85 | 99.30±0.95 | 99.27±0.64 | 95.15±6.72 | 92.33±4.45 | 99.18±0.83 | **99.91±0.11** | 96.76±1.36 |
| 13 | 91.50±5.44 | 97.08±0.78 | 86.41±20.06 | 96.48±2.50 | 93.13±7.02 | 79.50±20.39 | 99.43±0.46 | 94.91±1.86 | 91.70±6.44 | 94.29±4.42 | **98.57±1.26** | 97.67±1.90 |
| 14 | 94.95±5.19 | 81.65±16.19 | 91.79±2.16 | 92.71±1.65 | 93.11±6.86 | 94.37±2.32 | 85.75±12.31 | **95.02±3.38** | 74.27±20.48 | 75.08±12.59 | 93.01±4.23 | 90.50±5.08 |
| 15 | 83.54±2.56 | 89.69±1.84 | 83.35±2.77 | 77.63±2.50 | 90.42±5.92 | 84.83±2.72 | 89.40±3.19 | 87.62±3.93 | 86.37±4.81 | 84.23±1.63 | 83.90±1.66 | **96.00±2.17** |
| 16 | 98.19±0.99 | 96.42±1.77 | 98.40±0.17 | 74.77±1.88 | 86.32±5.98 | 98.04±1.38 | 98.60±0.27 | 98.39±2.16 | 96.27±2.82 | 99.04±0.32 | 98.83±0.07 | **99.58±1.11** |
| OA (%) | 94.76±0.72 | 95.19±0.65 | 94.96±0.48 | 88.17±0.52 | 94.90±0.74 | 95.01±0.34 | 95.33±0.24 | 95.11±0.68 | 90.92±0.75 | 93.90±0.44 | 94.43±0.15 | **96.96±0.27** |
| AA (%) | 96.06±0.93 | 96.01±1.39 | 96.03±1.40 | 89.62±0.44 | 95.71±0.56 | 95.41±0.90 | 96.20±0.72 | 95.69±0.57 | 92.46±2.08 | 95.10±0.93 | 96.43±0.25 | **97.63±0.34** |
| $\kappa \times 100$ | 94.16±0.80 | 94.65±0.72 | 94.39±0.54 | 86.85±0.57 | 94.32±0.83 | 94.44±0.38 | 94.02±0.26 | 94.55±0.76 | 89.91±0.83 | 93.21±0.49 | 93.80±0.17 | **96.61±0.30** |
| PR (%) | 97.15±0.56 | 97.60±0.20 | 97.09±0.21 | 89.57±1.66 | 95.86±0.58 | 97.55±0.15 | 96.95±0.27 | 95.79±0.59 | 89.40±1.79 | 96.68±0.41 | 97.13±0.20 | **97.86±0.12** |

are shown in Table VIII and Fig. 11, respectively. From the classification results, it can be seen that S2DBFT achieves superior OA, AA, Kappa, and PR coefficients of 95.72%, 93.11%, 94.32%, and 95.95%, which are 1.48%–14.59%, 2.43%–23.88%, 2.04%–19.88%, and 0.72%–17.85% higher than those of the other methods, respectively. Among the comparison methods, IRTS-3DCNN [53] and SF [39] yield poorer results. IRTS-3DCNN [53] mainly relies on IRTS-CNNs by integrating feedback systems during the iterative training process, which fails to capture subtle interclass differences in complex scenes. SF [39], on the other hand, uses the Transformer architecture to capture long-range dependencies but struggles to adapt to the specific spectral feature space, leading to suboptimal classification performance. This highlights that both CNN- and Transformer-based methods, during their early development stages, generally lack competitive classification performance. In contrast, S2DBFT integrates the strengths of both CNN and Transformer, offering enhanced

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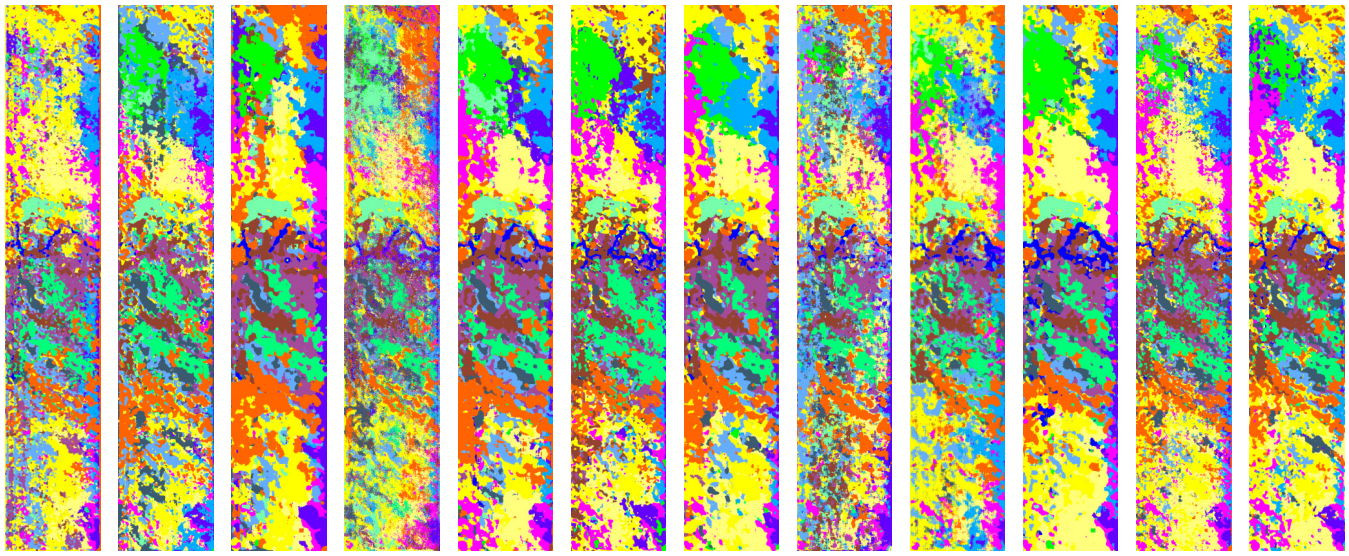



Fig. 13. Classification maps obtained by different methods in Botswana (2% for training). (a) IRTS-3DCNN (87.96%). (b) HybridSN (86.17%). (c) DBDA (88.72%). (d) SF (82.37%). (e) GAHT (93.39%). (f) SSFTT (91.66%). (g) SSTN (94.25%). (h) MorphFormer (85.23%). (i) SS-MTr (85.26%). (j) SQSFormer (92.96%). (k) SSEFN (94.43%). (l) S2DBFT (95.33%).

TABLE X

CLASSIFICATION ACCURACIES OF DIFFERENT METHODS IN BOTSWANA (2% FOR TRAINING)

| Class | IRTS-3DCNN [53] | HybridSN [34] | DBDA [35] | SF [39] | GAHT [49] | SSFTT [47] | SSTN [40] | morphformer [50] | SS-MTr [56] | SQSFormer [41] | SSEFN [51] | **S2DBFT (Ours)** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 92.27±5.72 | 97.58±4.85 | 94.39±4.65 | 95.83±3.68 | 95.91±3.84 | 94.32±4.84 | 95.30±5.15 | 72.58±6.58 | 95.83±4.31 | 100.00±0.00 | 100.00±0.00 | **100.00±0.00** |
| 2 | 49.90±15.84 | 20.40±6.71 | 76.16±29.67 | 60.00±15.85 | 53.94±18.68 | 77.37±24.80 | 81.82±13.51 | 75.76±22.25 | 76.97±15.64 | 63.03±11.04 | 95.15±1.96 | **95.34±10.33** |
| 3 | 94.72±4.28 | 98.70±0.98 | 93.25±3.70 | 83.33±5.78 | 98.94±1.45 | 95.77±3.49 | 97.32±3.22 | 94.47±4.20 | 93.17±6.20 | 93.82±5.22 | 98.54±0.41 | **99.02±1.95** |
| 4 | 99.53±0.95 | 96.78±6.45 | **100.00±0.00** | 95.07±5.72 | 99.81±0.38 | 99.81±0.38 | **100.00±0.00** | 98.86±2.27 | 99.15±1.14 | **100.00±0.00** | **100.00±0.00** | **100.00±0.00** |
| 5 | 79.92±6.28 | 72.58±5.13 | 79.85±3.91 | 68.64±10.89 | 85.53±4.17 | 81.97±5.04 | 84.32±4.87 | 76.82±4.48 | 62.50±9.60 | 88.26±2.21 | **90.23±0.37** | 85.08±2.40 |
| 6 | 80.38±13.89 | 78.11±10.64 | 68.79±5.79 | 49.70±15.10 | 86.89±5.52 | 85.61±5.84 | 90.30±4.76 | 69.77±11.35 | 55.00±9.49 | 91.29±3.84 | 87.65±1.69 | **100.00±0.00** |
| 7 | 93.39±6.37 | 95.91±6.44 | 99.84±0.31 | 94.57±6.02 | **100.00±0.00** | 94.41±4.19 | **100.00±0.00** | 93.15±8.23 | 96.85±3.24 | 99.06±1.38 | 99.84±0.31 | 88.34±10.55 |
| 8 | 74.97±4.77 | 70.55±5.49 | 75.38±8.05 | 66.33±9.19 | 75.58±4.94 | 76.68±8.32 | 73.97±3.79 | 67.04±16.48 | 79.10±10.16 | 83.22±9.95 | 87.74±3.43 | **89.16±8.56** |
| 9 | 84.81±7.15 | 83.44±12.80 | 93.90±8.20 | 91.30±5.61 | 97.92±2.67 | 84.74±6.36 | 98.25±1.81 | 79.09±12.06 | 81.56±4.47 | 90.39±7.65 | 78.51±2.13 | **99.51±0.99** |
| 10 | 94.49±5.40 | 96.79±3.63 | 67.82±19.44 | 90.95±4.73 | 97.37±5.27 | 98.68±1.09 | **100.00±0.00** | 97.70±3.11 | 85.60±9.79 | 97.37±1.99 | 99.67±0.40 | 99.60±0.65 |
| 11 | 97.79±2.16 | 98.19±3.61 | 99.80±0.27 | 96.79±3.14 | **100.00±0.00** | 99.53±0.94 | **100.00±0.00** | 98.33±2.70 | 98.39±1.13 | 99.33±0.63 | 99.93±0.13 | 99.98±0.04 |
| 12 | 92.09±11.07 | 85.54±12.07 | 98.64±1.98 | 88.70±4.52 | 99.55±0.66 | 97.63±2.49 | 97.18±2.11 | 89.04±8.46 | 85.65±8.12 | 87.12±14.66 | 98.42±0.42 | **100.00±0.00** |
| 13 | 96.65±2.55 | 98.48±2.86 | 96.96±6.08 | 88.37±5.93 | 99.62±0.76 | 99.47±0.89 | 99.62±0.59 | **99.70±0.61** | 93.46±6.15 | 97.64±1.47 | 98.78±0.81 | 99.47±0.46 |
| 14 | 58.92±14.14 | 44.09±22.54 | 87.96±6.88 | 42.15±17.38 | 83.66±1.72 | 85.16±5.83 | 82.80±0.00 | 58.49±10.01 | **88.82±5.99** | 83.44±4.28 | 86.45±1.87 | 83.87±1.18 |
| OA (%) | 87.96±2.04 | 86.17±0.74 | 88.72±3.86 | 82.37±1.67 | 93.39±1.25 | 91.66±1.58 | 94.25±1.24 | 85.23±1.98 | 85.26±1.40 | 92.96±0.64 | 94.43±0.52 | **95.33±0.84** |
| AA (%) | 84.99±3.18 | 81.22±1.51 | 88.05±4.70 | 79.41±2.75 | 91.05±1.70 | 90.80±2.21 | 92.92±1.15 | 83.63±2.56 | 85.15±1.09 | 91.00±0.69 | 94.35±0.58 | **94.90±0.80** |
| $\kappa \times 100$ | 86.94±2.23 | 84.98±0.81 | 87.76±4.19 | 80.87±1.83 | 92.82±1.36 | 90.96±1.72 | 93.76±1.35 | 83.98±2.15 | 84.03±1.52 | 92.37±0.70 | 93.96±0.57 | **94.94±0.91** |
| PR (%) | 89.98±1.25 | 89.04±1.66 | 91.39±0.89 | 83.17±2.28 | 94.52±0.45 | 95.38±0.36 | 95.06±0.21 | 87.16±2.01 | 86.70±2.64 | 93.61±0.51 | 94.92±0.40 | **95.40±0.38** |

spectral–spatial feature extraction capabilities, and excels particularly in classifying objects with complex backgrounds and high spectral similarity.

*2) Salinas:* On the Salinas dataset, the classification accuracy tables and maps for each method are shown in Table IX and Fig. 12, respectively. As shown in the table, S2DBFT outperforms all other methods with OA, AA, Kappa, and PR values exceeding 96%, specifically achieving 96.96%, 97.63%, 96.61%, and 97.86%, respectively. Additionally, S2DBFT achieves the optimal performance across all three metrics, with improvements of 1.63%–9.87%, 1.20%–8.01%, 1.96%–10.97%, and 0.26%–8.46% over the other methods. Among the 16 land-cover classes in this dataset, S2DBFT achieves the highest classification accuracy for four categories, while SSTN [40] demonstrates exceptional performance in mixed pixel classification, achieving perfect accuracy 100.00% for five land-cover classes. Traditional Transformer approaches SS-MTr [56] and SQSFormer [41] exhibit limitations in local information processing, yielding lower OA values 90.92% and 93.90%, respectively. Hybrid CNN–Transformer methods SSFTT [47] and MorphFormer [50] show improved performance, yet still fall short of S2DBFT's capabilities. Notably, Transformer-based methods generally outperform CNN-based approaches on this dataset, suggesting their superior capability in modeling complex spectral characteristics.

*3) Botswana:* On the Botswana dataset, the classification accuracy tables and maps for each method are shown in Table X and Fig. 13, respectively. Analysis of the results reveals that the proposed S2DBFT outperforms all other comparison methods and achieves the best performance across all three metrics-OA, AA, Kappa, and PR. With only 2% of the training samples, our S2DBFT model achieves an OA of 95.33%, an AA of 94.90%, a Kappa of 94.94%, and a PR of 95.40%, outperforming the other methods by 0.9%–16.47%, 0.55%–19.94%, 0.98%–17.92%, and 0.02%–12.23%, respectively. Despite the limited training samples and imbalanced class distribution, the S2DBFT model achieves superior classification results compared to the 11 other methods. Among the comparison algorithms, the hybrid model HybridSN [34], based on 2-D CNN and 3-D CNN, exhibits strong feature extraction capabilities but suffers from a low OA of 86.17% under the limited training sample scenario, significantly lower than the other methods. The DBDA model, by integrating an effective attention mechanism into both spectral and spatial feature extraction branches, shows significant improvement over HybridSN [34]. GAHT [49] achieves superior perfor-

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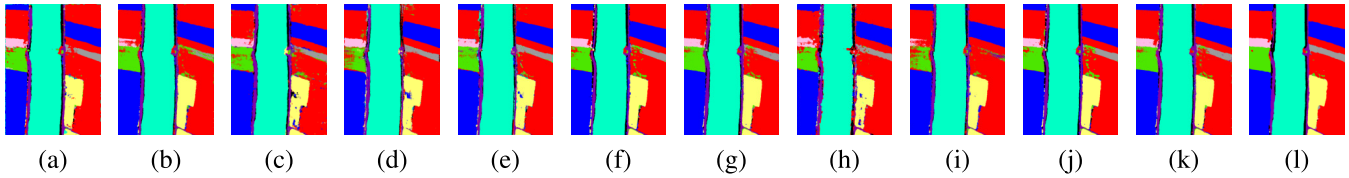



Fig. 14. Classification maps obtained by different methods in WHU-HI-LongKou (0.2% for training). (a) IRTS-3DCNN (97.68%). (b) HybridSN (96.45%). (c) DBDA (96.43%). (d) SF (96.25%). (e) GAHT (96.85%). (f) SSFTT (97.18%). (g) SSTN (97.23%). (h) MorphFormer (96.23%). (i) SS-MTr (93.91%). (j) SQSFormer (96.27%). (k) SSEFN (97.58%). (l) S2DBFT (98.64%).

TABLE XI

CLASSIFICATION ACCURACIES OF DIFFERENT METHODS IN WU-HI-LONGKOU (0.2% FOR TRAINING)

| Class | IRTS-3DCNN [53] | HybridSN [34] | DBDA [35] | SF [39] | GAHT [49] | SSFTT [47] | SSTN [40] | morphformer [50] | SS-MTr [56] | SQSFormer [41] | SSEFN [51] | **S2DBFT (Ours)** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 99.67±0.07 | 99.52±0.48 | 99.99±0.01 | 99.60±0.19 | 99.97±0.02 | 99.93±0.08 | 99.99±0.00 | 99.76±0.05 | 97.99±1.73 | 98.00±0.18 | 99.99±0.01 | **100.00±0.00** |
| 2 | 96.38±2.28 | 92.64±4.12 | 98.14±1.23 | 85.21±4.13 | 95.87±1.77 | **99.03±0.41** | 90.47±0.47 | 95.13±2.19 | 83.78±5.34 | 94.38±1.44 | 93.10±0.82 | 98.21±0.36 |
| 3 | 92.40±4.60 | 87.26±2.94 | 87.87±3.72 | 81.23±7.47 | 87.48±6.54 | 94.64±1.92 | 87.81±0.79 | 90.60±2.56 | 86.48±5.30 | 81.50±2.90 | 96.53±1.12 | **96.59±0.46** |
| 4 | 97.94±0.48 | 98.22±0.35 | 97.92±0.47 | 97.60±0.65 | 98.23±0.35 | 98.37±0.28 | 97.44±0.16 | **99.03±0.13** | 95.88±1.44 | 97.67±0.18 | 98.81±0.15 | 98.80±0.32 |
| 5 | 80.23±6.25 | 84.86±4.60 | 72.33±10.26 | 80.88±6.67 | 73.77±11.35 | 93.21±3.01 | 72.03±1.78 | 87.83±5.48 | 68.72±10.52 | 68.80±5.41 | 92.94±0.80 | **94.28±2.39** |
| 6 | 98.54±0.77 | 96.07±1.30 | 99.18±0.74 | 93.90±1.33 | 99.58±0.26 | 99.59±0.20 | 99.64±0.03 | 98.86±0.40 | 83.14±9.20 | **99.72±0.14** | 98.02±1.45 | 99.45±0.50 |
| 7 | 99.92±0.05 | 99.92±0.03 | 99.86±0.15 | **100.00±0.00** | 99.60±0.28 | **100.00±0.00** | 99.99±0.00 | 99.67±0.09 | 99.60±0.08 | 96.00±0.27 | 99.99±0.00 | 99.97±0.02 |
| 8 | 90.99±1.32 | 87.68±6.56 | 92.89±1.78 | 86.84±4.41 | 87.13±5.81 | **94.00±1.66** | 91.98±0.43 | 88.63±2.81 | 78.35±3.16 | 92.27±2.64 | 91.11±0.69 | 93.93±0.99 |
| 9 | 78.8±2.24 | 81.89±7.80 | 70.02±3.34 | 66.59±6.94 | 56.55±11.80 | 79.92±2.85 | 78.97±1.28 | 76.83±1.22 | 56.47±5.23 | 77.05±3.43 | 70.84±0.33 | **82.17±3.02** |
| OA (%) | 96.68±0.14 | 96.45±0.39 | 96.43±0.26 | 96.25±0.40 | 96.85±0.48 | 97.18±0.22 | 97.23±0.06 | 96.23±0.19 | 93.91±0.98 | 96.27±0.10 | 97.58±0.06 | **98.64±0.13** |
| AA (%) | 92.76±1.20 | 92.12±1.34 | 90.91±1.21 | 87.98±1.08 | 88.69±2.11 | 95.41±0.58 | 90.92±0.20 | 91.83±0.95 | 83.38±2.70 | 88.15±0.48 | 93.48±0.10 | **95.76±0.44** |
| Kappa ×100 | 96.95±0.19 | 95.64±0.55 | 96.62±0.34 | 95.05±0.53 | 95.85±0.64 | 97.00±0.16 | 96.35±0.07 | 96.57±0.33 | 91.97±1.31 | 95.41±0.14 | 97.04±0.08 | **98.21±0.17** |
| PR (%) | 95.20±0.33 | 96.38±0.22 | 96.51±0.19 | 93.44±0.58 | 93.42±0.61 | 96.40±0.20 | 96.25±0.17 | 94.15±0.46 | 78.87±2.86 | 95.71±0.26 | 95.56±0.35 | **96.53±0.24** |

mance via its GPE module and locally constrained MHSA mechanism. SSEFN [51] emerges as the strongest competitor through its spatial–spectral enhancement strategy and adaptive decision fusion module. S2DBFT particularly excels in classifying eight land-cover categories, achieving perfect accuracy of 100% for Classes 1, 4, 6, and 12.

*4) WHU-HI-LongKou:* On the WHU-Hi-LongKou dataset, the classification accuracy tables and maps for each method are shown in Table XI and Fig. 14, respectively. The results show that CNN- and Transformer-based methods exhibit similar classification performance on this dataset, with all algorithms achieving satisfactory classification performance. However, S2DBFT remains competitive, with an OA of 98.64%, an AA of 95.76%, a Kappa of 98.21%, and a PR of 96.53%. Its OA, AA, Kappa, and PR are 1.06%–6.27%, 0.35%–32.76%, 1.17%–10.85%, and 0.02%–17.66% higher than those of the other methods, respectively. Furthermore, the table shows that S2DBFT has the lowest standard deviation, indicating higher stability and consistency of its results. The SSEFN [51], by combining hybrid CNN and Transformer structures, enables the joint extraction of fine spectral–spatial features across different scenes, and, as the method most similar to S2DBFT, achieves an OA of 97.58%, demonstrating strong competitive performance in terms of individual class accuracy.

## V. DISCUSSION

This section begins by discussing the interclass feature separability of the S2DBFT method across the four datasets mentioned above and provides a comprehensive analysis through clustering visualization results. By comparing S2DBFT with the Transformer-based SQSFormer [41] and the CNN–Transformer hybrid SSEFN [51], as well as conducting ablation studies on each module, it is evident that S2DBFT demonstrates clearer class boundaries. In addition, the background issue is addressed through detailed quantitative experiments, validating the effectiveness of our proposed method S2DBFT in handling interference from background samples during training.

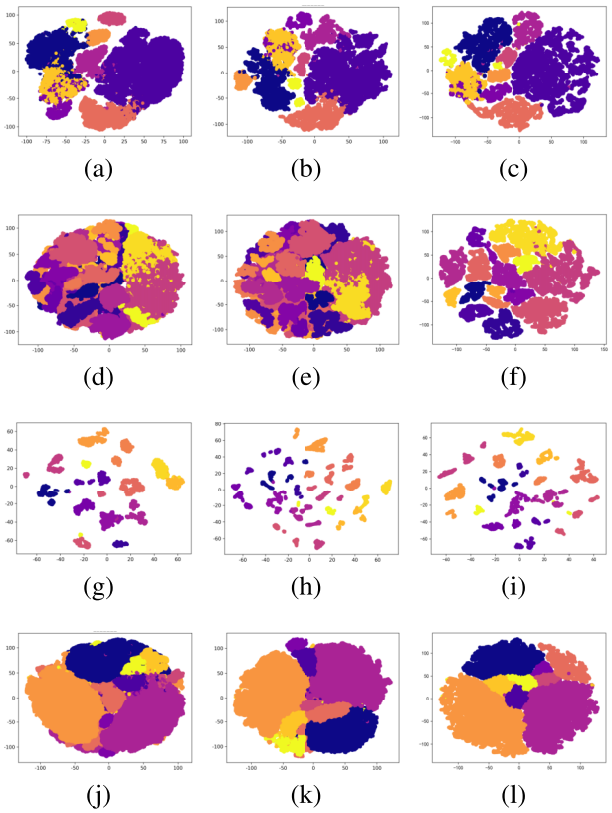



Fig. 15. Feature separability for different variants of different models. (a) SQSFormer. (b) SSEFN. (c) S2DBFT for University of Pavia. (d) SQSFormer. (e) SSEFN. (f) S2DBFT for Salinas. (g) SQSFormer. (h) SSEFN. (i) S2DBFT for Batswana. (j) SQSFormer. (k) SSEFN. (l) S2DBFT for WU-HI-LongKou.

### A. Feature Separability

In general, excellent feature separability is often characterized by the intraclass cohesion and the interclass separability in the feature space. To vividly illustrate the separability of the features of our proposed method, we selected a Transformer-based method, SQSFormer [41], and a CNN–Transformer hybrid method, SSEFN [51], to compare with our method

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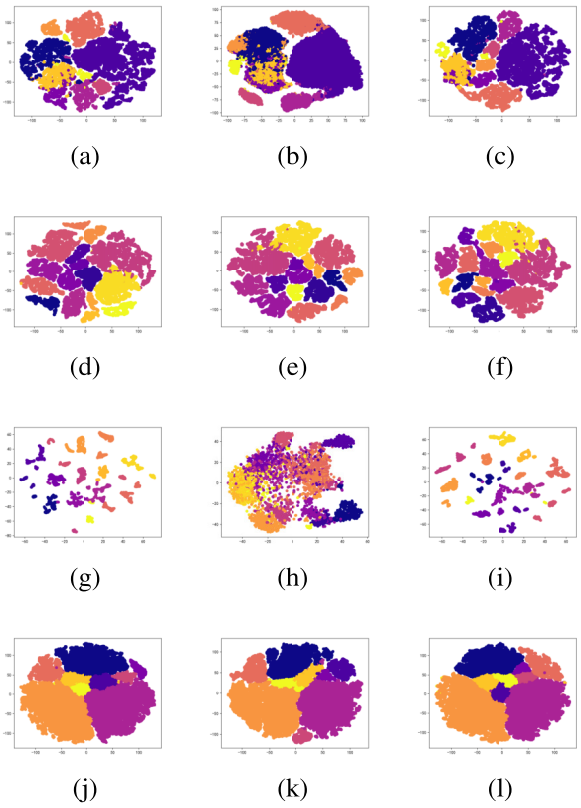



Fig. 16. Feature separability for different variants of S2DBFT. (a) Without SAM. (b) Without FSSM. (c) S2DBFT for University of Pavia. (d) Without SAM. (e) Without FSSM. (f) S2DBFT for Salinas. (g) Without SAM. (h) Without FSSM. (i) S2DBFT for Batswana. (j) Without SAM. (k) Without FSSM. (l) S2DBFT for WU-HI-LongKou.

using feature extraction and aggregation visualization techniques. Since these two models both adopt spectral spatial fusion methods, they are similar to our proposed S2DBFT. Additionally, we conducted an ablation study with cluster visualization analysis to further validate the effectiveness of the individual modules of our proposed method. As shown in Figs. 15 and 16, we performed experiments on four datasets: University of Pavia, Salinas, Botswana, and WHU-Hi-LongKou.

*1) For Comparative Analysis:* As shown in Fig. 15, the cluster visualization clearly shows that when compared to the other two methods across the four datasets, the proposed S2DBFT framework not only provides clearer class boundaries but also significantly reduces confusion between different classes and enhances the cohesion of samples within the same class. Specifically, on the University of Pavia dataset, our method shows noticeably better aggregation within each class, indicating its ability to effectively capture both spectral and spatial features in the image. This is particularly evident in distinguishing complex regions, such as roads and buildings, where our method outperforms the others. On the Salinas dataset, our clustering results also demonstrate a higher degree of class separability. Compared to the baseline methods, our approach is better at grouping similar land covers (e.g., different crop types) while effectively reducing misclassification within the same class. For the Botswana dataset, our clustering results exhibit greater separability, with less overlap between categories compared to the other two methods. On the WHU-Hi-LongKou dataset, the clustering results reveal that our method handles categories with high spectral similarity more effectively, with clearer boundaries in the clustering results.

*2) For Ablation Analysis:* Fig. 16 illustrates the clustering visualization results of the ablation study. In this study, we systematically removed the SPAEM and SPEEM modules from our proposed method, comparing the clustering outcomes under different ablation settings. The clustering charts from the ablation experiments reveal that the resolution and accuracy of the clustering results significantly degrade when either the FSSM or SAM module is removed, indicating that these components play a crucial role in the effectiveness of our approach. Specifically, for the University of Pavia dataset, the removal of any module leads to a noticeable drop in accuracy, with a marked increase in confusion between classes. For the Salinas dataset, the removal of any module causes significant overlap between Classes 4 and 5. For the Botswana dataset, compared to the removal of any individual module, the full method demonstrates a significantly higher classification performance for individual classes and a reduction in interclass confusion. Finally, in the case of the WHU-Hi-LongKou dataset, the removal of any module results in a reduction in the smoothness of the class boundaries in the clustering results, leading to partial overlap between classes.

### B. Background Classification

In conventional HSIC studies, model development and evaluation are typically guided by a prescribed GT map; all unlabeled pixels are relegated to a "background" (BKG) class and discarded from training and testing. Pixels are therefore labeled as BKG, not because they lack discriminative spectral content, but simply because reliable annotations are unavailable. Two factors account for the prevalence of such background samples: 1) the inherent spectral–spatial complexity of hyperspectral data renders exhaustive labeling prohibitively expensive and 2) modern sensors capture subtle material variations that cannot be readily identified by existing prior knowledge or visual inspection. To quantify a classifier's behavior when these pixels are retained, we report four BKG-aware metrics—$\mathrm{P_{OA}}$, $\mathrm{P_{AA}}$, $\mathrm{P_{OPR}}$, and $\mathrm{P_{APR}}$

$$P_{\mathrm{PR}}(C_m) = \text{precision rate } of \ C_m = \frac{\hat{n}_{mm}}{\hat{n}_m} \tag{34}$$

$$P_{\mathrm{OA}} = \frac{1}{N}\sum_{m=1}^{M} n_{mm} = \sum_{m=1}^{M} \frac{n_m}{N} P_A(C_m) \tag{35}$$

$$P_{\mathrm{AA}} = \frac{1}{M}\sum_{m=1}^{M} \frac{n_{mm}}{n_m} = \frac{1}{M}\sum_{m=1}^{M} P_A(C_m) \tag{36}$$

$$P_{\mathrm{OPR}} = \sum_{m=1}^{M} P\left(\hat{C}_m\right) P_{\mathrm{PR}}\left(\hat{C}_m\right) = \sum_{m=1}^{M} \frac{\hat{n}_m}{\hat{N}} P_{\mathrm{PR}}(C_m) \tag{37}$$

$$P_{\mathrm{APR}} = \frac{1}{M}\sum_{m=1}^{M} P_{\mathrm{PR}}\left(\hat{C}_m\right). \tag{38}$$

To rigorously assess the background robustness of the proposed S2DBFT, we conducted experiments on the University

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TABLE XII

$P_{PR}$, $P_{OA}$(BKG), $P_{AA}$(BKG), $P_{OPR}$(BKG), AND $P_{APR}$(BKG) OF DIFFERENT METHODS IN PU DATASET (0.5% FOR TRAINING)

| Class | IRTS-3DCNN [53] | HybridSN [34] | DBDA [35] | SF [39] | GAHT [49] | SSFTT [47] | SSTN [40] | morphformer [50] | SS-MTr [56] | SQSFormer [41] | SSEFN [51] | S2DBFT (Ours) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | $P_{PR}$ (%) | | | | | | |
| 1 | 56.43±3.44 | 48.57±1.91 | 49.02±1.76 | 39.58±4.28 | 55.32±1.23 | 54.79±2.29 | **61.22±3.13** | 55.83±2.86 | 18.32±4.92 | 22.77±4.41 | 48.89±2.65 | 49.16±2.74 |
| 2 | 78.94±0.89 | 74.89±2.15 | **89.03±0.39** | 80.42±0.73 | 83.37±0.08 | 87.06±0.55 | 81.26±1.42 | 72.04±2.92 | 68.40±3.90 | 69.30±1.82 | 68.40±2.51 | 79.93±0.34 |
| 3 | **66.15±1.47** | 32.70±4.58 | 50.39±2.80 | 40.77±3.58 | 60.36±2.32 | 58.73±3.44 | 65.26±1.71 | 61.50±1.45 | 27.17±4.65 | 31.78±3.06 | 49.38±3.06 | 49.39±3.99 |
| 4 | 44.68±4.92 | 37.70±3.00 | 37.88±2.79 | 43.46±4.36 | 39.93±0.78 | 30.44±4.52 | 43.87±4.86 | 37.21±4.10 | 19.18±4.43 | 23.42±4.74 | 46.00±2.32 | **57.88±4.01** |
| 5 | 78.54±1.85 | **84.04±0.69** | 67.08±3.88 | 79.87±1.09 | 71.15±1.28 | 74.14±0.09 | 75.83±1.19 | 72.57±3.66 | 63.27±1.36 | 58.06±2.33 | 66.70±1.05 | 72.88±2.11 |
| 6 | 73.00±1.28 | 73.04±2.01 | 80.15±0.33 | 68.54±2.45 | 77.15±0.59 | 78.61±0.81 | **88.13±0.23** | 74.83±0.96 | 37.44±3.40 | 33.78±4.07 | 70.24±0.95 | 80.44±1.22 |
| 7 | 81.07±0.84 | 76.04±0.81 | 62.90±0.93 | 56.73±4.67 | 58.52±3.29 | **84.04±0.04** | 69.75±2.32 | 66.01±1.18 | 18.71±5.73 | 38.06±3.35 | 68.24±1.76 | 63.40±2.29 |
| 8 | 62.30±4.60 | 49.43±2.47 | 57.69±3.80 | 51.50±3.79 | 51.55±3.75 | **66.79±1.81** | 59.49±2.59 | 62.79±1.05 | 14.68±4.50 | 20.85±4.88 | 49.56±2.35 | 44.44±3.14 |
| 9 | **75.00±1.33** | 39.11±1.14 | 34.20±4.52 | 29.41±6.02 | 38.94±4.55 | 20.00±3.94 | 31.84±3.97 | 34.47±3.48 | 6.34±6.97 | 13.64±5.94 | 44.12±4.71 | 57.66±2.99 |
| $P_{OA}$(BKG)(%) | 43.64±5.21 | 59.12±3.08 | 65.64±4.64 | 37.12±4.33 | 54.89±1.05 | 57.27±2.70 | 60.04±3.24 | 63.03±3.34 | 46.31±3.90 | 45.09±2.78 | 60.19±3.12 | **65.79±1.07** |
| $P_{AA}$(BKG)(%) | 34.48±4.65 | 46.73±2.64 | 53.73±3.01 | 23.94±4.01 | 45.33±3.33 | 47.11±1.73 | 49.91±3.13 | 51.93±3.88 | 28.09±4.86 | 36.08±0.80 | 52.77±2.57 | **53.78±1.26** |
| $P_{OPR}$(BKG)(%) | 86.44±0.77 | 85.69±0.40 | 86.91±1.01 | 83.76±0.26 | 87.92±0.39 | 87.66±0.23 | 87.36±0.22 | 88.54±1.70 | 79.31±0.30 | 84.50±1.24 | 85.47±0.74 | **88.74±0.63** |
| $P_{APR}$(BKG)(%) | 67.65±0.91 | 62.97±1.13 | 62.42±1.45 | 57.14±2.36 | 65.87±1.39 | 67.24±1.32 | 62.29±1.16 | 65.84±5.76 | 39.13±4.10 | 56.16±1.70 | 61.98±0.68 | **67.78±1.26** |

of Pavia scene, which contains extensive unlabeled areas, strong shadow effects, and intricate class-transition zones. All hyperparameters (training split, number of epochs, learning rate, and so on) follow the protocol described in Section V. Eleven recent competitors—IRTS-3DCNN [53], HybridSN [34], DDBDA [35], SF [39], SSFTT [49], GAHT [47], SSTN [40], MorphFormer [50], SS-MTr [56], SQSFormer [41], and SSEFN [51]—were first benchmarked under the conventional background-free setting and then reevaluated with BKG pixels included (HSIC-B) under identical conditions. As summarized in Table XII, several methods that excel under HSIC-NB experience a marked accuracy degradation once background is reinstated. By contrast, S2DBFT attains the best scores on $P_{OA}$(BKG), $P_{AA}$(BKG), $P_{OPR}$(BKG), and $P_{APR}$(BKG), outperforming the baselines by 0.15%–28.67%, 0.05%–29.84%, 0.20%–9.43%, and 0.13%–28.65%, respectively. In particular, S2DBFT improves $P_{OPR}$(BKG) by 3.27–4.24 percentage points over the latest SSEFN [51] and SQSFormer [41], indicating that the proposed spectral weighting and dual-branch interaction preserve fine-grained targets such as vegetation and bare soil more effectively. Although GAHT [47], SSFTT [49], and SSTN [40] achieve comparable OA, they lag behind S2DBFT in background discrimination performance and robustness. Overall, even without an explicit background-modeling module, the shallow spectral–spatial feature extractors, interaction-aware attention, and multiscale fusion strategy embedded in S2DBFT provide strong representation power for background regions, demonstrating excellent adaptability and discriminative capability in real-world, background-rich scenarios.

## VI. CONCLUSION

In this work, we propose the S2DBFT for HSI classification. PCA is employed to reduce the dimensionality of the HSI. The SPEEM and SPAEM modules are designed to extract shallow spectral and spatial features, respectively. The SAM module is developed to recalibrate the spectral features through reweighting. Additionally, the FSSM is introduced to incorporate the spectral weights from the SAM module into spatial feature weighting via MHS3A, enabling interactive fusion of spectral and spatial features. The spatial–spectral interactive features are then fused with the spectral features from the SAM output, followed by GAP and fully connected layers for classification. Extensive experiments on four benchmark HSI datasets demonstrate that S2DBFT achieves robust and superior classification performance, particularly under challenging conditions characterized by limited training samples and complex background settings.

Although the proposed S2DBFT has yielded promising results, future work will focus on enhancing the FSSM to further strengthen the information interaction capability between spectral and spatial features. This refinement aims to achieve even higher accuracy in HSI classification, especially in complex background scenarios.

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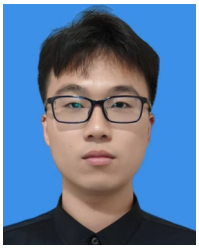
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